THE CLASSIFICATION BEHIND REMARK 3.11 OF [CGSS]

Abstract. We classify the nonempty indecomposable finite-type integer Cartan matrices and, using the base and root-preservation theorems in Mathlib, the indecomposable reduced crystallographic Root Systems. We give the complete weighted-graph argument, coordinate models with bases in Mathlib's index order, and the line counts and matrix equations of Remark 3.11 of [CGSS]. The same argument proves Smith's theorem and the assertion about deleting a vector after Theorem 3.8. We separate the library interface from the arguments proved here and record exact data for the finite verifications. We also record the nonempty hypothesis needed for matrices and the corrections needed in the printed statements.

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1. Introduction

Remark 3.11 of Cameron, Goethals, Seidel and Shult [CGSS] reduces the classification of Root Systems with unequal root lengths to an assertion that the remaining possibilities follow readily. We will supply a proof by bases and finite-type Cartan matrices. The combinatorial part is the weighted version of the argument for Smith's theorem in [BCN, Theorem 3.2.5]; we will write out the part that that proof leaves to inspection.

We use Mathlib [Mathlib], commit `082e2d37e8b0463410cdb532e111cd43d5a66174`. The matrix definitions, rather than their informal names in other sources, fix our indexing. In particular, the denominator in Mathlib's root pairing is the squared length of the *second* root.

Theorem A. Let I be a nonempty finite set and let $M : I \times I \longrightarrow \mathbb{Z}$ satisfy `Matrix.IsFiniteCartan` and `Matrix.IsIndecomposable`. There are a positive integer ℓ, a bijection $\phi : \{0, \ldots, \ell - 1\} \to I$, and exactly one type and parameter from the following list such that $M(\phi(i), \phi(j))$ is the (i, j)-entry of that named matrix for every i, j:

$$\begin{gathered} \mathrm{A}_\ell\ (\ell \geq 1), \qquad \mathrm{B}_\ell\ (\ell \geq 2), \qquad \mathrm{C}_\ell\ (\ell \geq 3), \qquad \mathrm{D}_\ell\ (\ell \geq 4), \\ \mathrm{E}_6,\ \mathrm{E}_7,\ \mathrm{E}_8,\ \mathrm{F}_4,\ \mathrm{G}_2. \end{gathered} \tag{1}$$

These are Mathlib's definitions `A`, `B`, `C`, `D`, `E`, $\mathtt{F}_4$ and $\mathtt{G}_2$ in the namespace `CartanMatrix`, respectively. The last two identifiers use Unicode subscript digits. Every matrix in this list has both properties.

"Exactly one" concerns the type and the parameter, not the bijection. This is a reindexing from the named matrix's index set to I. The empty matrix satisfies both Mathlib predicates. Thus the version of Theorem A with an arbitrary finite index set and no rank-zero alternative is false. Equivalently, for arbitrary I we must add the empty matrix to the list.

We will prove Theorem A, give its coordinate realizations, and use precisely the Mathlib results listed in Section 12 to obtain the Root System classification. We will then prove every line-system assertion in Remark 3.11, including its final example, and obtain Theorem 3.8 without using the classification of Root Systems. This supplies ordinary proofs for the structured proofs of the blueprint and for the Lean formalization; none of the Python checks is a premise of a proof.

2. Preliminaries and conventions

The purpose of this section is to establish notation and the complete hypotheses used below. All Euclidean spaces are finite-dimensional real inner product spaces. We will use functional notation for entries: $M(i, j)$ and $\mathbf{x}(i)$. The vectors $\mathbf{u}_1, \ldots, \mathbf{u}_m$ form the standard orthonormal basis of $\mathbb{R}^m$. Subscripts on these vectors index a family, not their entries. Dynkin types are upright. The letter C without an upright typeface will always denote the line-system matrix of [CGSS], not a Cartan type.

Definition 2.1. A *Root System* in a Euclidean space V is a finite subset $\Phi \subseteq V \setminus \{0\}$ which spans V and has the following properties:

(1) $2\langle\alpha, \beta\rangle / \langle\alpha, \alpha\rangle \in \mathbb{Z}$ for all $\alpha, \beta \in \Phi$;
(2) if $t \in \mathbb{R}$, $\alpha \in \Phi$ and $t\alpha \in \Phi$, then $t = 1$ or $t = -1$;
(3) for all $\alpha, \beta \in \Phi$,

$$s_\alpha(\beta) = \beta - \frac{2\langle\alpha, \beta\rangle}{\langle\alpha, \alpha\rangle}\alpha \in \Phi.$$

We will call a Root System *indecomposable* if it is nonempty and cannot be partitioned into two nonempty mutually orthogonal subsets. A *similarity* between Root Systems is a linear bijection f between their ambient spaces, with $f(\Phi) = \Psi$, for which some $a > 0$ satisfies $\langle f\mathbf{x}, f\mathbf{y}\rangle = a\langle\mathbf{x}, \mathbf{y}\rangle$ for all $\mathbf{x}, \mathbf{y}$. Thus $a^{-1/2} f$ is an isometry. This is what we mean by isometry after rescaling.

Reflection is an orthogonal involution: writing $\mathbf{x} = t\alpha + \mathbf{y}$ with $\langle \alpha, \mathbf{y} \rangle = 0$ gives $s_\alpha \mathbf{x} = -t\alpha + \mathbf{y}$. In particular, $-\alpha \in \Phi$ and each reflection permutes Φ. We allow the empty Root System in the zero space, but do not call it indecomposable. In Definition 3.9 of [CGSS], the denominator already presupposes nonzero roots; its condition (b) also excludes zero by taking $t = 0$ if zero were a root. We read its printed S in (b) as Σ.

For roots indexed by i, put

$$p(i,j) = \frac{2\langle \alpha_i, \alpha_j \rangle}{\langle \alpha_j, \alpha_j \rangle}. \tag{2}$$

This is Mathlib's `RootPairing.pairing`. A *base* in the usual root sense is a linearly independent subset Δ of roots such that every root is an integer combination of Δ with all coefficients nonnegative or all nonpositive. It spans V. Mathlib also includes the corresponding condition for coroots; `RootPairing.Base.mk'` supplies that condition from the root conditions, with the hypotheses in Section 12. The Cartan matrix of a base is $M(i,j) = p(i,j)$.

A finite integer matrix is *of finite Cartan type* when its diagonal is 2, its off-diagonal entries are nonpositive, its zero entries are symmetric, and there is a function $d : I \longrightarrow \mathbb{Z}_{>0}$ such that $H = \operatorname{diag}(d)M$ is symmetric and $\mathbf{x}^T H \mathbf{x} > 0$ for every nonzero integer vector $\mathbf{x}$. This is exactly `Matrix.IsFiniteCartan`. In `Matrix.PosDef` the vectors are finitely supported; on a finite index set this is the same condition. Hermitian over $\mathbb{Z}$ means symmetric. Mathlib's `Matrix.IsIndecomposable` says that a reindexing to $\begin{pmatrix} A & B \\ 0 & D \end{pmatrix}$ forces one diagonal block to have size zero.

A *weighted graph* is a finite nonempty vertex set with a symmetric real matrix B, zero diagonal, and nonnegative off-diagonal entries. Its *support* joins i to j when $B(i,j) > 0$; connectedness refers to this ordinary simple graph. An edge has weight $B(i,j)$. We require a connected graph to be nonempty. We will write $Y \preccurlyeq X$ if there is an injection $\phi : V(Y) \to V(X)$ such that $B_Y(u,v) \le B_X(\phi(u), \phi(v))$ for every $u, v \in V(Y)$. Edges may be omitted and weights may be decreased. The containment, with this injection, is *proper* if ϕ is not surjective or at least one of these inequalities is strict. For a real symmetric matrix B on a finite nonempty set, we will define

$$\lambda(B) = \max_{\langle \mathbf{x}, \mathbf{x} \rangle = 1} \mathbf{x}^T B \mathbf{x}, \qquad \lambda(X) = \lambda(B_X).$$

The maximum exists by continuity and compactness of the nonempty unit sphere. In Lemma 4.1 we will prove that it is an eigenvalue and bounds every real eigenvalue. We will use *largest eigenvalue* in this sense throughout.

3. From integer positivity to a weighted graph

Lemma 3.1. *Let H be a symmetric matrix on a finite set, with rational entries. If $\mathbf{x}^T H \mathbf{x} > 0$ for every nonzero integer vector $\mathbf{x}$, then $\mathbf{y}^T H \mathbf{y} > 0$ for every nonzero real vector $\mathbf{y}$.*

Proof. Multiplying H by a positive common denominator changes neither the hypothesis nor the conclusion, so we may assume its entries are integers. Clearing denominators of a nonzero rational vector proves strict positivity on rational vectors. Continuity of the quadratic polynomial and density of $\mathbb{Q}^I$ in $\mathbb{R}^I$ give nonnegativity on real vectors: a negative value would persist on a neighbourhood containing a rational vector.

Suppose $\mathbf{y} \neq 0$ and $\mathbf{y}^T H \mathbf{y} = 0$. For any real vector $\mathbf{z}$ and real t,

$$0 \le (\mathbf{y} + t\mathbf{z})^T H (\mathbf{y} + t\mathbf{z}) = 2t\mathbf{z}^T H \mathbf{y} + t^2 \mathbf{z}^T H \mathbf{z}.$$

If the coefficient of t were nonzero, choosing the opposite sign of t and sufficiently small absolute value would make the right side negative. Thus $\mathbf{z}^T H \mathbf{y} = 0$ for every $\mathbf{z}$, and $H\mathbf{y} = 0$. Since $\mathbf{y} \neq 0$, `Matrix.exists_mulVec_eq_zero_iff` over $\mathbb{R}$ gives $\det H = 0$ over $\mathbb{R}$. The determinant commutes

with the integer cast by `RingHom.map_det`, and this cast is injective, so $\det H = 0$ over $\mathbb{Z}$ as well. The same kernel theorem over $\mathbb{Z}$ gives a nonzero integer vector $\mathbf{v}$ with $H\mathbf{v} = 0$. Then $\mathbf{v}^T H\mathbf{v} = 0$, contrary to the hypothesis. This proves strict positivity over $\mathbb{R}$; Section 12 records the exact algebraic hypotheses. □

Lemma 3.2. *Let I be a nonempty finite set and let M be a matrix over a ring on I, with $M(i,j) = 0$ if and only if $M(j,i) = 0$. Its off-diagonal nonzero support is connected if and only if M is indecomposable in Mathlib's sense.*

Proof. A nonempty proper union of connected components has no entries to its complement in either direction. Enumerating the two sets gives a block diagonal decomposition with two positive block sizes. Conversely, a block-triangular decomposition with both block sizes positive has no entries from its second block to its first. Symmetry of the zero relation gives no entries in the other direction. A path cannot cross between the blocks, so the support is disconnected. □

Lemma 3.3. *Let I be nonempty and finite, and suppose $M : I \times I \to \mathbb{Z}$ is of finite Cartan type and is indecomposable. There is a connected weighted graph X on I whose edge weights belong to $\{1, \sqrt{2}, \sqrt{3}\}$, such that*

$$S = D^{1/2} M D^{-1/2} = 2I - B, \qquad B(i,j) = \sqrt{M(i,j)M(j,i)} \quad (i \neq j),$$

where $D = \operatorname{diag}(d)$ for any positive integer symmetrizer from the finite-type hypothesis. Moreover S is real symmetric positive definite and $\lambda(X) < 2$.

Proof. By Lemma 3.1, DM is positive definite over $\mathbb{R}$. Since $S = D^{-1/2}(DM)D^{-1/2}$, it is symmetric positive definite. Its diagonal is 2. If $i \neq j$, symmetry of DM gives $d(i)M(i,j) = d(j)M(j,i)$. Thus the square of $S(i,j)$ is $M(i,j)M(j,i)$ and $S(i,j) \leq 0$. Restrict the quadratic form to coordinates i, j. If $S(i,j) = b$, its value at $(1, -b/2)$ is $2 - b^2/2 > 0$. Thus $b^2 < 4$. The integer $M(i,j)M(j,i)$ is nonnegative, so it is 0, 1, 2 or 3. This proves the weight assertion. Connectedness follows from Lemma 3.2. For every unit real vector $\mathbf{x}$, $\mathbf{x}^T B\mathbf{x} = 2 - \mathbf{x}^T S\mathbf{x} < 2$. The maximum on the compact unit sphere is attained and is still strictly below 2. This maximum is $\lambda(X)$ by definition; we will identify it as an eigenvalue in Lemma 4.1. □

4. Rayleigh quotients and strict monotonicity

Lemma 4.1. *Let B be a real symmetric matrix on a finite nonempty set. The number*

$$\lambda = \max_{\langle \mathbf{x}, \mathbf{x} \rangle = 1} \mathbf{x}^T B\mathbf{x}$$

is an eigenvalue of B and bounds every real eigenvalue of B. A nonzero real vector has Rayleigh quotient λ if and only if it is a λ-eigenvector. If B is entrywise nonnegative and its support is connected, there is a strictly positive λ-eigenvector, and its λ-eigenspace is one-dimensional.

Proof. The quadratic function attains its maximum on the unit sphere. Let $\mathbf{x}$ attain it, and set $Q = \lambda I - B$. The definition of the maximum gives $\mathbf{y}^T Q\mathbf{y} \geq 0$ for every $\mathbf{y}$ and $\mathbf{x}^T Q\mathbf{x} = 0$. The polynomial argument in Lemma 3.1, now applied to Q, gives $Q\mathbf{x} = 0$, so the maximum is an eigenvalue. Every real eigenvalue is the Rayleigh quotient of a nonzero real eigenvector, and is thus at most λ. The same polynomial argument applies to any vector attaining the quotient; the converse follows by substitution. This is also the finite-dimensional Rayleigh fact provided by the declarations recorded in Section 12.

Suppose now that $B \geq 0$ and its support is connected. Replacing $\mathbf{x}$ entrywise by $|\mathbf{x}|$ does not decrease the numerator and does not change the denominator. Thus a nonnegative maximizing eigenvector exists. If one of its entries is zero, the eigenvector equation at that vertex is a sum of nonnegative terms equal to zero. Every neighbour also has zero entry. Propagation along paths would give the zero vector. Thus all entries are positive; call this vector $\mathbf{w}$.

If $\mathbf{v}$ is any other real λ-eigenvector, let $t = \max_i \mathbf{v}(i)/\mathbf{w}(i)$. Then $t\mathbf{w} - \mathbf{v}$ is a nonnegative eigenvector with at least one zero entry. The preceding propagation proves it is zero. Thus $\mathbf{v} = t\mathbf{w}$, proving that the real eigenspace is one-dimensional. □

Lemma 4.2. [BCN, Theorem 3.1.1 and Proposition 3.2.4] *Let X be a connected weighted graph with nonnegative weights and let Y be a nonempty weighted graph with nonnegative weights. If $Y \preccurlyeq X$, then $\lambda(Y) \le \lambda(X)$, with strict inequality whenever the containment is proper. Also, a positive eigenvector of a connected nonnegative symmetric matrix has eigenvalue equal to its largest eigenvalue.*

Proof. Choose a nonnegative unit maximizing vector for Y by taking absolute values in the Rayleigh argument; connectedness of Y is not required. Extend it by zero to the vertex set of X. Entrywise domination gives a Rayleigh quotient for X at least $\lambda(Y)$. If equality holds between the two largest eigenvalues, this vector is a nonnegative maximizing eigenvector of X and is positive by propagation. Thus there is no omitted vertex. If an entry was strictly increased, its contribution to the quadratic form, on positive coordinates, is strictly positive. There can be no such entry either.

For the last assertion, let $B\mathbf{v} = \theta\mathbf{v}$ with $\mathbf{v} > 0$, and let $B\mathbf{w} = \lambda\mathbf{w}$ with $\mathbf{w} > 0$ supplied by Lemma 4.1. Symmetry gives $(\lambda - \theta)\langle \mathbf{w}, \mathbf{v}\rangle = 0$. The inner product is positive, so $\lambda = \theta$. □

5. The affine obstructions and the complete dichotomy

We will specify the few weighted graphs that the argument uses. A path's weight sequence lists edges in their path order. Let $T(a, b, c)$ be the unweighted tree with a single vertex of degree 3 and arms of a, b, c edges, where $a, b, c \ge 1$.

Definition 5.1. Our *affine obstruction list* consists of the following graphs:

(1) the unweighted cycles Cyc_m, $m \ge 3$;
(2) $\widetilde{\mathrm{D}}_4 = K_{1,4}$, and, for $\ell \ge 5$, $\widetilde{\mathrm{D}}_\ell$: a path of $\ell - 4$ edges with two new leaves at each endpoint;
(3) $\widetilde{\mathrm{E}}_6 = T(2,2,2)$, $\widetilde{\mathrm{E}}_7 = T(1,3,3)$ and $\widetilde{\mathrm{E}}_8 = T(1,2,5)$;
(4) X_m^{end}, $m \ge 3$: a path on m vertices, with weight $\sqrt{2}$ at its first and last edges and weight 1 at every other edge;
(5) X_k^{fork}, $k \ge 1$: vertices $v_0, \dots, v_k$ on a path, with two new unit-weight leaves at v_0, and path weights 1 except that $v_{k-1}v_k$ has weight $\sqrt{2}$;
(6) X^{mid}: the path with weights $1, 1, \sqrt{2}, 1$;
(7) X^{triple}: the path with weights $\sqrt{3}, 1$.

For X_1^{fork} the edge at the branching vertex itself has weight $\sqrt{2}$. For X_3^{end} both edges have weight $\sqrt{2}$. These conventions include the small cases needed below.

Lemma 5.2. *Every graph in Definition 5.1 has largest eigenvalue 2.*

Proof. The positive vectors in Table 1 satisfy $B\mathbf{x} = 2\mathbf{x}$. We will give the local verification here, so that the infinite families are not reduced to finitely many experimental checks. On a cycle each entry is 1 and has two neighbours. On a $\widetilde{\mathrm{D}}$ graph the nonleaf path entries are 2 and the leaves have entry 1; a path interior has neighbour sum 4, a fork endpoint has sum $2 + 1 + 1 = 4$, and a leaf has neighbour sum 2. In $K_{1,4}$ the centre has entry 2 and neighbour sum 4.

On an arm of length a, put $\mathbf{x}(t) = h(a + 1 - t)/(a + 1)$ at distance t from a centre of value h. The entries form an arithmetic progression ending at a virtual zero after the leaf, so every noncentral neighbour sum is twice the entry. The central equation is

$$\frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1} = 1.$$

For the three triples this is $1/3+1/3+1/3=1$, $1/2+1/4+1/4=1$, and $1/2+1/3+1/6=1$.

In X_m^{end} the endpoints have entry 1 and all interior entries have value $\sqrt{2}$. Each end contributes $\sqrt{2}$ to its neighbouring interior vertex, just as another interior vertex does through a unit edge. The endpoint equation is $\sqrt{2}\sqrt{2}=2$. This also covers X_3^{end}. In X_k^{fork}, entries at $v_0,\dots,v_{k-1}$ are 2, at v_k the entry is $\sqrt{2}$, and both extra leaves have entry 1. The weighted neighbour contribution towards the terminal edge is $\sqrt{2}\sqrt{2}=2$; at the terminal vertex it is $2\sqrt{2}$. The centre receives $1+1+2=4$, including when $k=1$. The remaining path equations are $2+2=4$. For X^{mid}, the entries $1,2,3,2\sqrt{2},\sqrt{2}$ give neighbour sums $2,4,6,4\sqrt{2},2\sqrt{2}$ in that order. For X^{triple}, the entries $\sqrt{3},2,1$ give $2\sqrt{3},4,2$. All entries are positive, so Lemma 4.2 proves the assertion. □

Lemma 5.3. *Let X be a nonempty finite connected acyclic simple graph on m vertices, with every degree at most 2. There is a bijection $\phi:\{0,\dots,m-1\}\to V(X)$ such that, for all i,j,*

$$\phi(i)\sim\phi(j)\quad\Longleftrightarrow\quad |i-j|=1.$$

If a vertex w of degree at most 1 is prescribed, we may choose $\phi(0)=w$.

Proof. Choose a simple path $v_0,\dots,v_k$ of maximum length; such a path exists because X is finite and nonempty. A neighbour of either endpoint outside this path would extend it, contrary to maximality. An interior vertex already has its two distinct path neighbours, so the degree bound excludes any neighbour outside the path. Thus no edge leaves the set of path vertices. If a vertex lay outside that set, connectedness would give a path from it to v_0, and the first edge entering the set would be an edge leaving it. Thus the path visits every vertex. Its vertices are distinct, so $k=m-1$ and $\phi(i)=v_i$ is a bijection.

Every consecutive pair is adjacent. If $i<j$ and $j-i\geq 2$, an edge v_iv_j together with the path segment $v_i,\dots,v_j$ would form a cycle. Thus there are no other edges, proving the equivalence. When $m=1$, the unique vertex is $\phi(0)$. When $m\geq 2$, each interior vertex has degree 2 and both endpoints have degree 1. A prescribed vertex of degree at most 1 is thus an endpoint; reversing the bijection if necessary puts it at 0. □

For positive integers a,b,c, we will use the following labelled vertex set for $T(a,b,c)$:

$$V_{a,b,c}=\{*\}\ \sqcup\ (\{1\}\times\{1,\dots,a\})\ \sqcup\ (\{2\}\times\{1,\dots,b\})\ \sqcup\ (\{3\}\times\{1,\dots,c\}).$$

Its edges join $*$ to $(r,1)$ for $r=1,2,3$, and join consecutive positions on each arm. These are all its edges.

Lemma 5.4. *Let X be a finite connected acyclic simple graph with a vertex v of degree 3, all of whose other vertices have degree at most 2. There are positive integers a,b,c and a bijection $\phi:V_{a,b,c}\to V(X)$ with $\phi(*)=v$ whose edges are exactly*

$$\{v,\phi(r,1)\}\quad(r=1,2,3),\qquad \{\phi(r,t),\phi(r,t+1)\}\quad\text{for consecutive positions on arm } r.$$

We may choose $a\leq b\leq c$ by permuting the three arms.

Proof. There is a unique simple path between any two vertices of X. Connectedness gives a walk, from which deleting repeated portions gives a simple path. If two such paths were distinct, take their first divergence and their first subsequent reunion; the two intervening segments would form a cycle. This proves uniqueness.

Enumerate the three neighbours of v as w_1,w_2,w_3. For each such neighbour w, let X_w be the set of vertices other than v whose unique path from v starts with the edge vw. Each set contains w, and these three sets partition $V(X)\setminus\{v\}$. Every vertex on the segment from w to a vertex of X_w lies in X_w, by uniqueness of the path from v. Thus X_w induces a connected acyclic graph. Paths from v into distinct sets meet only at v: a later common vertex would have two different paths from v. An edge between distinct sets, together with those paths, would give a

cycle. There are no such edges. Also, w is the only neighbour of v in X_w, since the unique path to any neighbour of v is its single incident edge. Thus the three induced graphs are exactly the components after deleting v.

Their degrees are at most 2. Within X_w, the degree of w is at most 1, since deleting v removes one of its at most two neighbours. Apply Lemma 5.3 to each induced graph with starting vertex w. If its size is a_r, write the resulting ordered vertices as $\phi(r,1),\dots,\phi(r,a_r)$, with $\phi(r,1)=w_r$. All a_r are positive. Set $a=a_1, b=a_2, c=a_3$ and $\phi(*)=v$. The partition proves that ϕ is bijective; the path labellings, the absence of edges between the sets and the unique edge from each set to v give exactly the asserted edges. Permuting the three neighbours and their entire labelled arms orders the lengths as required. □

Lemma 5.5. *Let X be a finite connected simple graph containing none of Cyc_m, $\widetilde{\mathrm{D}}_\ell$, or $\widetilde{\mathrm{E}}_6, \widetilde{\mathrm{E}}_7, \widetilde{\mathrm{E}}_8$ as a subgraph. Then X admits a path labelling as in Lemma 5.3, or an arm labelling as in Lemma 5.4 of $T(1,1,c)$ with $c \geq 1$, or of one of $T(1,2,2)$, $T(1,2,3)$, $T(1,2,4)$.*

Proof. There is no cycle, so connectedness makes X a tree. A vertex of degree at least 4 and four of its neighbours would give $K_{1,4}$. Thus every degree is at most 3. If two vertices have degree 3, choose a pair at minimum distance. The internal vertices on their joining path have no degree 3, since any such vertex would give a closer pair. At each endpoint choose its two neighbours off the path. These four neighbours are distinct and are not on the path, since X is a tree. The selected path and four edges form $\widetilde{\mathrm{D}}_\ell$, where ℓ is the path length plus 4. Thus there is at most one degree-3 vertex.

If there is no degree-3 vertex, Lemma 5.3 gives the path bijection. Otherwise Lemma 5.4 gives $\phi : V_{a,b,c} \to V(X)$, with $1 \leq a \leq b \leq c$. Restricting this bijection to initial positions of each arm gives every truncation used next, with its edges preserved. If $a \geq 2$, truncating all three arms to length 2 gives $\widetilde{\mathrm{E}}_6$. Thus $a = 1$. If $b \geq 3$, truncating the second and third arms to length 3 gives $\widetilde{\mathrm{E}}_7$. Thus $b = 1$ or $b = 2$. The first case gives $T(1,1,c)$. In the second case, $c \geq 5$ would give $\widetilde{\mathrm{E}}_8$ by truncation; the remaining integers are $c = 2, 3, 4$. □

Theorem 5.6. *Let X be a finite connected weighted graph with edge weights in $\{1, \sqrt{2}, \sqrt{3}\}$. Then exactly one of the following holds:*

1. *X contains an affine obstruction in the sense of $\preccurlyeq$;*
2. *X is one of the following weighted Dynkin graphs: a unit path A_ℓ ($\ell \geq 1$); a path with one terminal $\sqrt{2}$ edge BC_ℓ ($\ell \geq 2$); $T(1,1,\ell-3) = \mathrm{D}_\ell$ ($\ell \geq 4$); $T(1,2,\ell-4) = \mathrm{E}_\ell$ ($\ell = 6,7,8$); the path $1, \sqrt{2}, 1$ of type F_4; or the single $\sqrt{3}$ edge of type G_2.*

Every graph in the second alternative has largest eigenvalue less than 2. In that alternative the identification comes with a path or arm bijection as in Lemmas 5.3 and 5.4, preserving the displayed edge weights.

Proof. Suppose no obstruction is contained in X. Lowering every weight to 1 and applying Lemma 5.5 shows in particular that the support has one of its explicit path or arm labellings.

If an edge of weight $\sqrt{3}$ is present and there is a third vertex, connectedness gives an edge at one endpoint of that edge leading towards a third vertex. Lowering the latter weight to 1 gives X^{triple}. Thus a triple edge forces the entire graph to be G_2. We may now assume all weights are 1 or $\sqrt{2}$.

Suppose a fork and a $\sqrt{2}$ edge are both present. There is just one fork. Choose its arm labelling ϕ and permute the arms to put the multiple edge on arm 1, ending at position $k \geq 1$. Map the path vertices $v_0,\dots,v_k$ of X_k^{fork} to $\phi(*), \phi(1,1),\dots,\phi(1,k)$, and map its extra leaves to $\phi(2,1), \phi(3,1)$. Thus we take the edge of each other arm at the fork. These vertices are distinct by bijectivity of ϕ. Lower all selected weights to 1 except the last edge on arm 1. The resulting

injection realizes $X_k^{\mathrm{fork}} \preccurlyeq X$. Thus no multiple edge can occur in a graph with a fork. Its unit weights now give one of the D or E cases of Lemma 5.5.

It remains to use a path labelling $\phi : \{0, \dots, \ell-1\} \to V(X)$. If two edges have weight $\sqrt{2}$, write them as $\{\phi(r), \phi(r+1)\}$ and $\{\phi(s), \phi(s+1)\}$ with $r < s$. The map $i \mapsto \phi(r+i)$ for $0 \le i \le s+1-r$ embeds X_{s-r+2}^{end}, after lowering the interior weights to 1. Thus there is at most one multiple edge. With none, the graph is A_ℓ. With one, reverse ϕ if necessary so that the numbers p, q of vertices on the two sides satisfy $1 \le p \le q$, with the p-vertex side first. The multiple edge is then $\{\phi(p-1), \phi(p)\}$ and $\ell = p+q$. If $p = 1$, the graph is BC_{p+q}. If $p = q = 2$, its weights are $1, \sqrt{2}, 1$, giving F_4. Every remaining case has $p \ge 2$ and $q \ge 3$. The five distinct vertices $\phi(p+2), \phi(p+1), \phi(p), \phi(p-1), \phi(p-2)$ are in the index range and give the weights $1, 1, \sqrt{2}, 1$ in that order. This is an explicit embedding of X^{mid}. This exhausts the cases.

For completeness, each listed Dynkin graph is a proper subgraph of an obstruction. A path with at least two vertices is obtained by deleting a vertex of the cycle on one more vertex; a single vertex is a subgraph of Cyc_3. The graph D_4 is $K_{1,4}$ with one leaf deleted. For $\ell \ge 5$, deleting one leaf at a fork endpoint of $\widetilde{\mathrm{D}}_\ell$ gives arms $1, 1, \ell-3$ at the other endpoint. For E_6 shorten one arm of $T(2,2,2)$ by one; for E_7 shorten a length-3 arm of $T(1,3,3)$ by one; for E_8 shorten the length-5 arm of $T(1,2,5)$ by one. A graph BC_ℓ is $X_{\ell+1}^{\mathrm{end}}$ with an endpoint deleted. Deleting the first endpoint of X^{mid} gives F_4, and deleting the unit-edge endpoint of X^{triple} gives G_2. Lemmas 5.2 and 4.2 give largest eigenvalue less than 2 in every case. Such a graph cannot contain an obstruction, again by monotonicity. Thus the two alternatives are exclusive. □

This determines precisely the obstruction list used by the proof. We do not require a classification of affine Cartan matrices, and do not attach an orientation or a Lie-theoretic affine name to $X_m^{\mathrm{end}}, X_k^{\mathrm{fork}}, X^{\mathrm{mid}}, X^{\mathrm{triple}}$.

6. Orientations, Mathlib's indices, and Theorem A

All matrix indices in this section are $0, \dots, \ell-1$, namely Mathlib's `Fin` convention. Unspecified off-diagonal entries are zero. The diagonal is always 2. The following formulas reproduce the definitions:

- A_ℓ: consecutive indices have both entries -1.
- B_ℓ: the same path, except $M(\ell-2, \ell-1) = -2$, $M(\ell-1, \ell-2) = -1$.
- C_ℓ: the transpose of B_ℓ.
- D_ℓ: unit edges $\{i, i+1\}$ for $0 \le i \le \ell-3$, and $\{\ell-3, \ell-1\}$. Equivalently, the fork has leaves $\ell-2, \ell-1$ and its other arm runs back to 0.
- E_ℓ: edges $\{0,2\}, \{1,3\}$ and $\{i, i+1\}$ for $2 \le i \le \ell-2$, with both entries -1.
- The last two matrices are

$$\mathrm{F}_4 = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -2 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}, \qquad \mathrm{G}_2 = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}.$$

The D_ℓ formula includes the consecutive edge $\{\ell-3, \ell-2\}$ and is exactly `CartanMatrix.D` for $\ell \ge 3$.

Proof of Theorem A. Apply Lemma 3.3 and Theorem 5.6. An obstruction would force largest eigenvalue at least 2, so the support with its weights is one of the listed Dynkin graphs. On a unit edge, the two negative integers have product 1, so both are -1. On a $\sqrt{2}$ edge they are $(-1,-2)$ in one of the two orders; on a $\sqrt{3}$ edge they are $(-1,-3)$ in one of the two orders. There is at most one multiple edge.

We will construct the bijection required in Theorem A from the labellings retained in Theorem 5.6. On a unit path, take the bijection of Lemma 5.3 as it stands. For $T(1,1,\ell-3)$ use the arm bijection ϕ and define $e:\{0,\dots,\ell-1\}\to I$ by

$$e(\ell-3)=\phi(*),\qquad e(\ell-2)=\phi(1,1),\qquad e(\ell-1)=\phi(2,1),$$
$$e(i)=\phi(3,\ell-3-i)\quad(0\le i\le \ell-4).$$

For $T(1,2,\ell-4)$ define

$$e(3)=\phi(*),\qquad e(1)=\phi(1,1),\qquad e(2)=\phi(2,1),\qquad e(0)=\phi(2,2),$$
$$e(i)=\phi(3,i-3)\quad(4\le i\le \ell-1).$$

Each formula uses every labelled arm position exactly once. Thus both maps are bijections, and the edge equivalences in the recognition lemmas give precisely the displayed A, D, E entries after reindexing. For a terminal double edge, reverse the path bijection if necessary to put its leaf at index $\ell-1$. The two entry orders give B_ℓ and C_ℓ. At rank 2, interchanging the two vertices interchanges these matrices, so we keep only B_2. Reversing the four-vertex path interchanges the two orientations in F_4. Interchanging the vertices does the same for G_2. Thus the list is exhaustive with Mathlib's exact indexing.

Conversely, use the positive integer symmetrizers

(3)

type	$(d(0),\dots,d(\ell-1))$
A, D, E	$(1,\dots,1)$
B	$(1,\dots,1,2)$
C	$(2,\dots,2,1)$
F_4	$(1,1,2,2)$
G_2	$(1,3)$.

For each edge the equality $d(i)M(i,j)=d(j)M(j,i)$ follows from its displayed entries; for a nonedge both sides vanish. The resulting $D^{1/2}MD^{-1/2}$ is $2I-B$ for the corresponding Dynkin graph. Its Rayleigh quotients are positive by Theorem 5.6. Congruence gives positivity of DM, including on nonzero integer vectors. All remaining finite-type conditions follow from the entries. The nonzero support is connected, so Lemma 3.2 gives indecomposability.

Here are explicit invariants for uniqueness. Rank is invariant. The products $M(i,j)M(j,i)$ determine the weighted support. A triple edge distinguishes G_2. A double edge distinguishes the remaining multiple-edge types from the simply laced types. In a path with at least three vertices, its being terminal distinguishes BC_ℓ from F_4, whose double edge is internal. In BC_ℓ for $\ell\ge 3$, the leaf incident with the double edge is intrinsic. The entry from its neighbour into it is -2 in B_ℓ and -1 in C_ℓ. Thus these are not permutation-equivalent. Among unit graphs, paths distinguish A. In a branched tree the unordered arm lengths distinguish D, which has at least two length-1 arms, from E, which has exactly one; rank distinguishes the three E cases. The rank-2 double-edge case has already been identified. These invariants prove exactly one type and parameter. □

Remark 6.1. The small-rank coincidences are matrix statements. Direct substitution in Mathlib's definitions gives

$$\mathrm{B}_1=\mathrm{C}_1=\mathrm{D}_1=\mathrm{A}_1=[2],\qquad \mathrm{D}_2=\begin{pmatrix}2&0\\0&2\end{pmatrix}.$$

Interchanging indices 0 and 1 takes B_2 to C_2. In D_3, reordering the old indices as $(1,0,2)$ gives A_3. The generalized definition E_ℓ has $\mathrm{E}_1=\mathrm{A}_1$, $\mathrm{E}_2=2I_2$, $\mathrm{E}_3\cong\mathrm{A}_2\oplus\mathrm{A}_1$ in order $(0,2,1)$, $\mathrm{E}_4\cong\mathrm{A}_4$ in order $(0,2,3,1)$, and $\mathrm{E}_5\cong\mathrm{D}_5$ in order $(0,2,3,1,4)$. For $\ell=9$ its graph is $T(1,2,5)$, so it is not of finite type; every $\ell>9$ contains this graph. All five definitions at parameter 0 give the empty matrix. The restrictions in (1) remove every coincidence and every decomposable positive-rank

small case. The docstring for `CartanMatrix.A` says "rank $n-1$", but its actual definition is a matrix on `Fin n`; we use rank n for that definition.

7. Smith's theorem and the deletion assertion

Corollary D. [CGSS, Theorem 3.8]; [BCN, Theorem 3.2.5] Let X be a finite connected simple graph. Its adjacency matrix has largest eigenvalue 2 if and only if X is a cycle Cyc_m ($m \geq 3$), $\widetilde{\mathrm{D}}_\ell$ ($\ell \geq 4$), or one of $\widetilde{\mathrm{E}}_6, \widetilde{\mathrm{E}}_7, \widetilde{\mathrm{E}}_8$. For each such graph on m vertices, $2I - B$ is the Gram matrix of spanning vectors of norm $\sqrt{2}$ in dimension $m-1$. Deleting any one vector leaves a basis of that space.

Proof. All obstruction weights in a graph with unit weights must themselves be 1. Thus Theorem 5.6 says that X either contains one of the unweighted affine graphs or is a unit Dynkin graph. The latter has largest eigenvalue less than 2. Proper containment of an affine graph has largest eigenvalue greater than 2 by Lemma 4.2. The affine graphs themselves have largest eigenvalue 2 by Lemma 5.2.

For the final assertion choose the positive vector $\mathbf{w}$ satisfying $B\mathbf{w} = 2\mathbf{w}$. For every real vector $\mathbf{x}$,

$$\mathbf{x}^T(2I - B)\mathbf{x} = \sum_{\{i,j\} \in E(X)} \mathbf{w}(i)\mathbf{w}(j) \left(\frac{\mathbf{x}(i)}{\mathbf{w}(i)} - \frac{\mathbf{x}(j)}{\mathbf{w}(j)} \right)^2. \tag{4}$$

Expanding gives the off-diagonal terms $-2\mathbf{x}(i)\mathbf{x}(j)$ and the coefficient $\sum_{j \sim i} \mathbf{w}(j)/\mathbf{w}(i) = 2$ on $\mathbf{x}(i)^2$. The form is positive semidefinite. Connectedness shows its kernel is exactly $\mathbb{R}\mathbf{w}$, since equality forces all displayed ratios to be equal. On the quotient $\mathbb{R}^m/\mathbb{R}\mathbf{w}$, this form is a positive definite inner product. The classes of the standard coordinate vectors have Gram matrix $2I - B$ and span a space of dimension $m-1$. Choosing an orthonormal basis identifies it with $\mathbb{R}^{m-1}$. Each vector has norm $\sqrt{2}$; adjacent vectors have inner product -1 and other distinct vectors have inner product zero. A relation omitting coordinate i has coefficient vector in $\mathbb{R}\mathbf{w}$ with a zero entry. Since $\mathbf{w}(i) > 0$, that relation is zero. Thus the remaining $m-1$ vectors are a basis. This also applies to any realization with this Gram matrix, since the norm of a linear combination is the same quadratic form. □

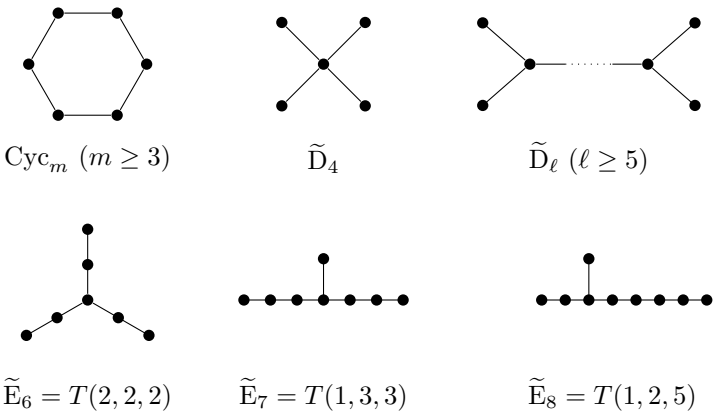


Figure 1. The graphs in Corollary D. The first drawing represents any cycle; the dotted segment in the $\widetilde{\mathrm{D}}$ drawing represents a path, whose two endpoints are distinct. All edges have weight 1.

8. Coordinate Root Systems and their bases

We will give coordinates with the standard Euclidean inner product. In each case the ambient space is the span specified below, rather than a larger coordinate space. Root bases have indices starting at 0; coordinate vectors $\mathbf{u}_i$ have indices starting at 1. Every sign in a set of the form $\{\pm\mathbf{u}_i \pm \mathbf{u}_j\}$ is chosen independently. Unqualified coordinate indices range over the coordinates of the stated ambient space.

8.1. **The classical systems.** For the ranges in (1), define

$$\begin{aligned}
\Phi(\mathrm{A}_\ell) &= \{\mathbf{u}_i - \mathbf{u}_j : 1 \le i \ne j \le \ell+1\}, & V(\mathrm{A}_\ell) &= \{\mathbf{x} : \textstyle\sum_i \mathbf{x}(i) = 0\},\\
\Phi(\mathrm{B}_\ell) &= \{\pm\mathbf{u}_i : 1 \le i \le \ell\} \cup \{\pm\mathbf{u}_i \pm \mathbf{u}_j : i < j\}, & V(\mathrm{B}_\ell) &= \mathbb{R}^\ell,\\
\Phi(\mathrm{C}_\ell) &= \{\pm 2\mathbf{u}_i : 1 \le i \le \ell\} \cup \{\pm\mathbf{u}_i \pm \mathbf{u}_j : i < j\}, & V(\mathrm{C}_\ell) &= \mathbb{R}^\ell,\\
\Phi(\mathrm{D}_\ell) &= \{\pm\mathbf{u}_i \pm \mathbf{u}_j : 1 \le i < j \le \ell\}, & V(\mathrm{D}_\ell) &= \mathbb{R}^\ell.
\end{aligned}$$

We also use the displayed C model at rank 2 when discussing its coincidence with B_2. The proposed bases are

$$\begin{aligned}
&\mathrm{A}_\ell: && \alpha_i = \mathbf{u}_{i+1} - \mathbf{u}_{i+2} && (0 \le i < \ell);\\
&\mathrm{B}_\ell: && \alpha_i = \mathbf{u}_{i+1} - \mathbf{u}_{i+2} && (0 \le i < \ell-1), && \alpha_{\ell-1} = \mathbf{u}_\ell;\\
&\mathrm{C}_\ell: && \alpha_i = \mathbf{u}_{i+1} - \mathbf{u}_{i+2} && (0 \le i < \ell-1), && \alpha_{\ell-1} = 2\mathbf{u}_\ell;\\
&\mathrm{D}_\ell: && \alpha_i = \mathbf{u}_{i+1} - \mathbf{u}_{i+2} && (0 \le i < \ell-1), && \alpha_{\ell-1} = \mathbf{u}_{\ell-1} + \mathbf{u}_\ell.
\end{aligned} \tag{5}$$

Lemma 8.1. *For $\ell \ge 1$ in type* A*, $\ell \ge 2$ in types* B, C*, and $\ell \ge 4$ in type* D*, the displayed sets are Root Systems in the displayed spaces. The vectors in* (5) *are bases, and their Cartan matrices are exactly Mathlib's matrices of the same type.*

Proof. All the sets are finite and have no zero. In each set, two vectors on one line differ by sign: their supports and the equal absolute values on a two-coordinate support determine this, while an axis has only its displayed pair of roots. For a root $\mathbf{u}_i - \mathbf{u}_j$, reflection interchanges coordinates i, j. For $\mathbf{u}_i + \mathbf{u}_j$, it interchanges those coordinates and changes both signs. For an axis root, reflection changes the sign of that coordinate, independent of whether the root is $\mathbf{u}_i$ or $2\mathbf{u}_i$. Negating a root leaves its reflection unchanged. Coordinate permutations preserve the A set and its zero-sum space. Signed coordinate permutations preserve each of the B, C, D sets. These descriptions verify closure under every root reflection.

The pairing coefficient with a root of squared length 2 is its inner product with the other root, an integer. In type B, a root of squared length 1 gives twice an integer coordinate. In type C, pairing against $\pm 2\mathbf{u}_i$, whose squared length is 4, gives $\pm\mathbf{x}(i)$, also an integer. Thus the crystallographic condition holds.

We will verify the base property by displaying the positive roots. Every root not on these lists is the negative of one on the lists. Empty sums below have value zero. In all four types, for $i < j$,

$$\mathbf{u}_i - \mathbf{u}_j = \sum_{k=i-1}^{j-2} \alpha_k.$$

In type A these exhaust one root from each pair of signs. In type B the remaining choices are

$$\mathbf{u}_i = \sum_{k=i-1}^{\ell-1} \alpha_k, \qquad \mathbf{u}_i + \mathbf{u}_j = \sum_{k=i-1}^{j-2} \alpha_k + 2\sum_{k=j-1}^{\ell-1} \alpha_k.$$

In type C they are

$$2\mathbf{u}_i = 2\sum_{k=i-1}^{\ell-2}\alpha_k + \alpha_{\ell-1}, \qquad \mathbf{u}_i + \mathbf{u}_j = \sum_{k=i-1}^{j-2}\alpha_k + 2\sum_{k=j-1}^{\ell-2}\alpha_k + \alpha_{\ell-1}.$$

In type D, for $i < j \le \ell - 1$, they are

$$\mathbf{u}_i + \mathbf{u}_j = \sum_{k=i-1}^{j-2}\alpha_k + 2\sum_{k=j-1}^{\ell-3}\alpha_k + \alpha_{\ell-2} + \alpha_{\ell-1},$$

and for $i < \ell$ they are

$$\mathbf{u}_i + \mathbf{u}_\ell = \sum_{k=i-1}^{\ell-3}\alpha_k + \alpha_{\ell-1}.$$

Each formula follows by telescoping the consecutive differences. For $\mathbf{u}_i + \mathbf{u}_j$ with $j \le \ell - 1$, the identity $\alpha_{\ell-2} + \alpha_{\ell-1} = 2\mathbf{u}_{\ell-1}$ supplies the endpoint. For $\mathbf{u}_i + \mathbf{u}_\ell$, we use only $\alpha_{\ell-1} = \mathbf{u}_{\ell-1} + \mathbf{u}_\ell$. These are nonnegative integer expansions of every indicated root.

The A roots span the zero-sum space: a vector in that space is $\sum_{i=1}^{\ell}\mathbf{x}(i)(\mathbf{u}_i - \mathbf{u}_{\ell+1})$. The axis roots span the B and C spaces. The D roots span since, for $i \ne j$, $2\mathbf{u}_i = (\mathbf{u}_i + \mathbf{u}_j) + (\mathbf{u}_i - \mathbf{u}_j)$. The expansions prove that the ℓ displayed simple roots span these ℓ-dimensional spaces, so they are independent. Thus they are bases in the root sense.

Finally, consecutive differences have squared norm 2 and inner product -1; nonconsecutive differences have disjoint support. In type B the last squared norm is 1 and its inner product with its predecessor is -1, giving entries $-2, -1$ in that order. In type C the last squared norm is 4 and that inner product is -2, giving entries $-1, -2$. In type D the final sum has squared norm 2, is orthogonal to $\mathbf{u}_{\ell-1} - \mathbf{u}_\ell$, and has inner product -1 with $\mathbf{u}_{\ell-2} - \mathbf{u}_{\ell-1}$; its other inner products are zero. Substitution in (2) gives every entry specified in Section 6. □

8.2. **The three** E **systems.** Define the lattice

$$L = \{\mathbf{x} \in \mathbb{Z}^8 : \textstyle\sum_i \mathbf{x}(i) \in 2\mathbb{Z}\} \ \cup\ \{\mathbf{x} \in (\mathbb{Z} + \tfrac{1}{2})^8 : \textstyle\sum_i \mathbf{x}(i) \in 2\mathbb{Z}\}. \tag{6}$$

Here "lattice" will mean an additive subgroup generated by a real basis; we will prove that assertion for L below. Put

$$\begin{aligned}
\Phi(\mathrm{E}_8) &= \{\pm\mathbf{u}_i \pm \mathbf{u}_j : i < j\} \ \cup\ \{\tfrac{1}{2}\textstyle\sum_{i=1}^{8}\epsilon(i)\mathbf{u}_i : \epsilon(i) \in \{1, -1\},\ \prod_i \epsilon(i) = 1\},\\
V(\mathrm{E}_7) &= \{\mathbf{x} \in \mathbb{R}^8 : \mathbf{x}(7) + \mathbf{x}(8) = 0\},\\
V(\mathrm{E}_6) &= \{\mathbf{x} \in \mathbb{R}^8 : \mathbf{x}(6) = \mathbf{x}(7) = -\mathbf{x}(8)\},\\
\Phi(\mathrm{E}_j) &= \Phi(\mathrm{E}_8) \cap V(\mathrm{E}_j) \qquad (j = 6, 7), \qquad V(\mathrm{E}_8) = \mathbb{R}^8.
\end{aligned}$$

The eight simple roots, with Mathlib's zero-based ordering, are

$$\begin{aligned}
\alpha_0 &= \tfrac{1}{2}(\mathbf{u}_1 + \mathbf{u}_8 - \mathbf{u}_2 - \mathbf{u}_3 - \mathbf{u}_4 - \mathbf{u}_5 - \mathbf{u}_6 - \mathbf{u}_7),\\
\alpha_1 &= \mathbf{u}_1 + \mathbf{u}_2, \qquad \alpha_i = \mathbf{u}_i - \mathbf{u}_{i-1} \quad (2 \le i \le 7).
\end{aligned} \tag{7}$$

The notation in the last formula uses coordinate indices: it gives $\alpha_2 = \mathbf{u}_2 - \mathbf{u}_1$, $\alpha_3 = \mathbf{u}_3 - \mathbf{u}_2$, up to $\alpha_7 = \mathbf{u}_7 - \mathbf{u}_6$. We use the first j vectors for type E_j.

Lemma 8.2. *The set L in (6) is an additive subgroup. Inner products of its elements are integers, and squared norms are even integers. Its vectors of squared norm 2 are exactly $\Phi(\mathrm{E}_8)$.*

Proof. Write $L_0 = \{\mathbf{x} \in \mathbb{Z}^8 : \sum_i \mathbf{x}(i) \text{ is even}\}$ and $\mathbf{v} = \frac{1}{2}\sum_i \mathbf{u}_i$. Then $L = L_0 \cup (L_0 + \mathbf{v})$ and $2\mathbf{v} \in L_0$. Thus addition and negation preserve L. For $\mathbf{x}, \mathbf{y} \in L_0$, $\langle \mathbf{x}, \mathbf{y}\rangle$ is an integer and

$\langle \mathbf{x}, \mathbf{v} \rangle = \sum_i \mathbf{x}(i)/2$ is an integer; also $\langle \mathbf{v}, \mathbf{v} \rangle = 2$. Expanding inner products proves integrality on both cosets. For $\mathbf{x} \in L_0$, $\langle \mathbf{x}, \mathbf{x} \rangle \equiv \sum_i \mathbf{x}(i) \equiv 0 \pmod 2$. For $\mathbf{x} + \mathbf{v} \in L_0 + \mathbf{v}$,

$$\langle \mathbf{x} + \mathbf{v}, \mathbf{x} + \mathbf{v} \rangle = \sum_{i=1}^{8} \mathbf{x}(i)(\mathbf{x}(i) + 1) + 2 \in 2\mathbb{Z}.$$

A nonzero lattice vector thus has squared norm at least 2. An integer vector of norm squared 2 has exactly two nonzero coordinates, both ± 1. A vector in the half-integer coset has all eight absolute coordinates at least $1/2$. Equality at norm squared 2 forces all to be $1/2$. Its coordinate sum is $4 - t$, where t is the number of negative signs; the sum is even precisely when t is even. This is exactly the displayed list of roots. □

Lemma 8.3. *Let L be an additive subgroup of a Euclidean space, every nonzero element of which has squared norm at least $b > 0$. Let $\alpha_0, \ldots, \alpha_{m-1}$ be a real basis generating L over $\mathbb{Z}$, with $\langle \alpha_i, \alpha_j \rangle \le 0$ for $i \ne j$. Every element of L of squared norm b has either all nonnegative or all nonpositive integer coefficients in this basis.*

Proof. Split its integer coefficient vector into its positive and negative parts, writing the element as $\mathbf{v} - \mathbf{w}$, where both parts are nonnegative combinations of disjoint subsets of the basis. Then $\langle \mathbf{v}, \mathbf{w} \rangle \le 0$. If both parts were nonzero, we would have

$$b = \langle \mathbf{v} - \mathbf{w}, \mathbf{v} - \mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{v} \rangle + \langle \mathbf{w}, \mathbf{w} \rangle - 2\langle \mathbf{v}, \mathbf{w} \rangle \ge 2b,$$

a contradiction. Thus one part is zero. □

Lemma 8.4. *The sets $\Phi(\mathrm{E}_j)$, $j = 6, 7, 8$, are Root Systems in their specified spaces. Their bases are the first j vectors in (7), and their Cartan matrices are exactly* `CartanMatrix.E j`.

Proof. All the listed simple roots lie in L and have squared norm 2. Given $\mathbf{x} \in \mathbb{R}^8$, its coefficients in (7) are

$$\begin{aligned} c(0) &= 2\mathbf{x}(8), \qquad c(1) = \tfrac{1}{2}\sum_{i=1}^{8} \mathbf{x}(i) + 2\mathbf{x}(8), \\ c(2) &= c(1) + \mathbf{x}(8) - \mathbf{x}(1), \\ c(k) &= \sum_{i=k}^{7} \mathbf{x}(i) + (8-k)\mathbf{x}(8) \quad (3 \le k \le 7). \end{aligned} \tag{8}$$

Substitution gives coordinate 8 equal to $c(0)/2$, coordinate 1 equal to $c(0)/2 + c(1) - c(2)$, coordinate 2 equal to $-c(0)/2 + c(1) + c(2) - c(3)$, coordinates $3 \le i \le 6$ equal to $-c(0)/2 + c(i) - c(i+1)$, and coordinate 7 equal to $-c(0)/2 + c(7)$. These are exactly the coordinates of $\mathbf{x}$, proving the inversion formula and linear independence. If $\mathbf{x} \in L$, then $c(0), c(1)$ are integers, $\mathbf{x}(8) - \mathbf{x}(1)$ is an integer, and each summand $\mathbf{x}(i) + \mathbf{x}(8)$ in $c(k)$ is an integer. Thus all coefficients are integers, so these roots generate L.

The conditions for $V(\mathrm{E}_7)$ and $V(\mathrm{E}_6)$ are respectively $c(7) = 0$ and $c(7) = c(6) = 0$. Thus their lattice intersections are generated by the first seven and the first six simple roots. They also span those subspaces, which have dimensions 7 and 6. Direct inner products in (7) give diagonal 2 and inner product -1 exactly on $\{0,2\}, \{1,3\}, \{2,3\}, \{3,4\}, \ldots, \{6,7\}$. To check the exceptional first vector, its coordinates at 1 and 8 are $1/2$ and those at 2 through 7 are $-1/2$, so its inner product with α_1 is zero, with α_2 is -1, and with $\alpha_3, \ldots, \alpha_7$ is zero. The remaining products follow from the two-coordinate supports of the displayed differences and sum. Thus all off-diagonal inner products are nonpositive, and Lemma 8.3, with $b = 2$, proves the root base property for L and for each sublattice intersection.

For a root α of norm squared 2, reflection takes $\mathbf{x} \in L$ to $\mathbf{x} - \langle \mathbf{x}, \alpha \rangle \alpha \in L$, since the inner product is an integer. Reflection preserves the norm, so it permutes the norm-2 vectors. If both vectors lie in one of the specified subspaces, their reflected vector remains there. Thus all three sets are closed under reflections and are crystallographic. Equal nonzero lengths give reducedness, and their simple roots prove spanning. Finally, denominator 2 in (2) makes the Cartan entries exactly the inner products just calculated, which are Mathlib's E_j entries. □

8.3. **The F_4 system.** In $\mathbb{R}^4$ put

(9)
$$\Phi(\mathrm{F}_4) = \{\pm\mathbf{u}_i \pm \mathbf{u}_j : i < j\} \cup \{\pm\mathbf{u}_i : 1 \le i \le 4\} \cup \{\tfrac{1}{2}(\epsilon(1)\mathbf{u}_1 + \epsilon(2)\mathbf{u}_2 + \epsilon(3)\mathbf{u}_3 + \epsilon(4)\mathbf{u}_4) : \epsilon(i) = \pm 1\}.$$

Its proposed ordered base is

$$\alpha_0 = \mathbf{u}_2 - \mathbf{u}_3, \quad \alpha_1 = \mathbf{u}_3 - \mathbf{u}_4, \quad \alpha_2 = \mathbf{u}_4, \quad \alpha_3 = \tfrac{1}{2}(\mathbf{u}_1 - \mathbf{u}_2 - \mathbf{u}_3 - \mathbf{u}_4). \tag{10}$$

Lemma 8.5. *The set (9) is a Root System with base (10) and Cartan matrix `CartanMatrix.F₄`. Its first two simple roots are long and its last two are short.*

Proof. The squared lengths are 2 on the two-coordinate roots and 1 on the other roots. Supports show there are no proportional roots except sign pairs; the axes show spanning. Signed coordinate permutations preserve the set. Reflections in the two-coordinate and axis roots are signed coordinate permutations as in Lemma 8.1.

It remains to check a half-coordinate root. Every such root is carried to $\mathbf{v} = (\mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_3 + \mathbf{u}_4)/2$ by a diagonal sign change preserving the set. An orthogonal transformation ϕ satisfies $s_{\phi\alpha} = \phi s_\alpha \phi^{-1}$, by substitution in the reflection formula. Thus it suffices to check $s_{\mathbf{v}}$ on all three kinds of roots. Its formula is

$$s_{\mathbf{v}}\mathbf{x} = \mathbf{x} - \left(\sum_i \mathbf{x}(i)\right)\mathbf{v}.$$

An axis vector $\mathbf{u}_i$ goes to $\mathbf{u}_i - \mathbf{v}$, a half-coordinate root; its negative goes to the negative of that image. A two-coordinate root with opposite signs is fixed. One with equal sign ϵ on coordinates i, j goes to $-\epsilon(\mathbf{u}_k + \mathbf{u}_l)$ on the other two coordinates. For a half-coordinate root with t plus signs, the coordinate sum is $t - 2$. For $t = 0$ or 4 its image is its negative; for $t = 2$ it is fixed; for $t = 1$ its image is the positive axis at its unique plus sign; for $t = 3$ its image is the negative axis at its unique minus sign. This proves closure under every reflection.

Pairing against a long root is an inner product. With a half-coordinate root this is a sum of two numbers $\pm 1/2$, and with an integer root it is an integer. Pairing against an axis root is twice a coordinate and is an integer. Pairing against a half-coordinate root, of squared length 1, is twice the inner product. For an integer root that is a signed sum of integer coordinates. For another half-coordinate root it is $(4 - 2t)/2 = 2 - t$, where t counts differing signs. Thus all pairings are integers.

The coefficients of a vector $\mathbf{x}$ in the ordered base are

$$\big(\mathbf{x}(1) + \mathbf{x}(2), \quad 2\mathbf{x}(1) + \mathbf{x}(2) + \mathbf{x}(3), \quad 3\mathbf{x}(1) + \mathbf{x}(2) + \mathbf{x}(3) + \mathbf{x}(4), \quad 2\mathbf{x}(1)\big). \tag{11}$$

Substitution in (10) proves this inversion formula and independence. Orient a root with $\mathbf{x}(1) \neq 0$ so that $\mathbf{x}(1) > 0$. If it is a half-coordinate root, these four coefficients are integers in $[0,1], [0,2], [0,3], \{1\}$, respectively. If $\mathbf{x}(1) = 1$, the other coordinates have at most one nonzero entry, equal to 1 or -1. The coefficients are again nonnegative integers. If $\mathbf{x}(1) = 0$, orient the first nonzero coordinate among 2, 3, 4 to be positive. There are at most two nonzero coordinates, each of absolute value 1. Every prefix sum of these three coordinates is nonnegative: after any initial zeros it starts with 1 and can subsequently decrease by at most 1. The coefficients in (11) are those prefix sums and zero. This proves the base property for every root.

The simple squared lengths are $(2, 2, 1, 1)$ and the only nonzero off-diagonal inner products are $\langle \alpha_0, \alpha_1 \rangle = -1$, $\langle \alpha_1, \alpha_2 \rangle = -1$ and $\langle \alpha_2, \alpha_3 \rangle = -1/2$. Equation (2) gives exactly the displayed F_4 matrix. □

8.4. **The G_2 system.** In the plane $V = \{\mathbf{x} \in \mathbb{R}^3 : \mathbf{x}(1) + \mathbf{x}(2) + \mathbf{x}(3) = 0\}$ put

$$\Phi(G_2) = \{\mathbf{u}_i - \mathbf{u}_j : i \neq j\} \cup \{\pm(2\mathbf{u}_i - \mathbf{u}_j - \mathbf{u}_k) : \{i, j, k\} = \{1, 2, 3\}\}, \tag{12}$$

$$\alpha_0 = \mathbf{u}_1 - 2\mathbf{u}_2 + \mathbf{u}_3, \qquad \alpha_1 = \mathbf{u}_2 - \mathbf{u}_3. \tag{13}$$

Lemma 8.6. *The set (12) is a Root System in V, with ordered base (13) and Cartan matrix `CartanMatrix.G₂`. Simple root 0 is long and simple root 1 is short.*

Proof. The six difference roots have squared length 2 and the six other roots have squared length 6. Their coordinate patterns give six distinct sign pairs, so they are reduced. Difference roots span V. Permuting coordinates and negating all coordinates preserve the set. A short-root reflection interchanges two coordinates. For the long root $\beta = 2\mathbf{u}_1 - \mathbf{u}_2 - \mathbf{u}_3$, a vector $\mathbf{x} \in V$ has $\langle \mathbf{x}, \beta \rangle = 3\mathbf{x}(1)$, so

$$s_\beta \mathbf{x} = (-\mathbf{x}(1), \mathbf{x}(2) + \mathbf{x}(1), \mathbf{x}(3) + \mathbf{x}(1)) = (-\mathbf{x}(1), -\mathbf{x}(3), -\mathbf{x}(2)).$$

This preserves the set. Each other long root is a coordinate permutation of β or its negative; conjugating by that permutation proves closure for each of them. A pairing against a short root is an integer inner product, and a pairing against β is $\mathbf{x}(1)$, an integer on the root set. Permutation and sign give integrality against every long root.

In the order (α_0, α_1), the positive coefficient pairs are

$$(0, 1),\ (1, 1),\ (1, 2),\ (1, 0),\ (1, 3),\ (2, 3). \tag{14}$$

The first three vectors are, in order, $\mathbf{u}_2 - \mathbf{u}_3, \mathbf{u}_1 - \mathbf{u}_2, \mathbf{u}_1 - \mathbf{u}_3$. The last three are $\mathbf{u}_1 - 2\mathbf{u}_2 + \mathbf{u}_3, \mathbf{u}_1 + \mathbf{u}_2 - 2\mathbf{u}_3, 2\mathbf{u}_1 - \mathbf{u}_2 - \mathbf{u}_3$. Together with their negatives these are all twelve roots. The simple Gram matrix is $\begin{pmatrix} 6 & -3 \\ -3 & 2 \end{pmatrix}$, whose determinant is 3, so the two roots are independent. Formula (14) proves the base property, and dividing its columns by the respective half squared lengths gives $\begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$, as required. □

Lemma 8.7. *Every coordinate Root System constructed in this section, in the parameter ranges stated for its construction, is indecomposable.*

Proof. The calculated simple-root nonorthogonality graph is connected in every case. If all roots were partitioned into two orthogonal nonempty subsets, each pair of nonorthogonal simple roots would lie in the same part. Following paths puts all simple roots in one part. Every root in the other part would be perpendicular to their spanning set, and would be zero, a contradiction. This includes a one-element base, since it already spans. □

9. From Root Systems to Mathlib and back

Lemma 9.1. *Let Φ be a Root System in a Euclidean space V. It defines a finite, crystallographic, reduced root pairing over $\mathbb{R}$ which is a root system in Mathlib's sense. If Φ is nonempty, its indecomposability is equivalent to Mathlib's irreducibility.*

Proof. Use the finite index set $\iota = \Phi$, the two spaces V, V, and the bilinear form $\texttt{inner}_\text{l}\, V$ for their inner product (the identifier is `inner` followed by Unicode subscript l). It is injective in each argument: if a difference $\mathbf{z}$ pairs to zero with every vector, pair it with $\mathbf{z}$ to obtain $\langle \mathbf{z}, \mathbf{z} \rangle = 0$, so

$\mathbf{z} = 0$. Finite dimensionality and `LinearMap.IsPerfPair.of_injective` make it a perfect pairing. The root embedding is inclusion and the coroot embedding sends α to

$$\alpha^\vee = \frac{2\alpha}{\langle \alpha, \alpha \rangle}.$$

This map is injective: applying the same operation twice returns the original nonzero vector. Its pairing with its root is 2. Let the reflection permutation on ι be s_α restricted to Φ; it is a permutation since it is an involution. The root reflection identity is the defining formula. For the coroot identity, orthogonality of s_α gives

$$(s_\alpha \beta)^\vee = s_\alpha(\beta^\vee) = \beta^\vee - \langle \alpha, \beta^\vee \rangle \alpha^\vee.$$

These are exactly the two identities in the structure constructor `RootPairing`. Its pairing is (2); condition (a) of Definition 2.1, with the two roots interchanged, supplies `IsCrystallographic`. A dependent pair of nonzero vectors over $\mathbb{R}$ is proportional; condition (b) supplies `IsReduced`. Roots span by assumption and their nonzero scalar multiples, the coroots, have the same span. Thus `IsRootSystem` holds.

We will check irreducibility explicitly. Suppose W is a subspace invariant under every s_α. If some $\mathbf{w} \in W$ has $\langle \mathbf{w}, \alpha \rangle \neq 0$, then $\mathbf{w} - s_\alpha \mathbf{w}$ is a nonzero multiple of α, so $\alpha \in W$. Otherwise $\alpha \in W^\perp$. Thus every root lies in W or $W^\perp$. If $W \neq 0$ and $W \neq V$, both sets of roots are nonempty: if all roots lay in one, their span would force the other subspace to be zero. They are disjoint since a nonzero vector cannot belong to $W \cap W^\perp$, and they are mutually orthogonal. This contradicts indecomposability. The root and coroot reflections on our two copies of V are the same orthogonal maps, so both irreducibility conditions hold. Nonemptiness of Φ gives $V \neq 0$, as required by Mathlib's definition. Alternatively, `RootPairing.IsIrreducible.mk'` supplies the coroot half of this argument.

Conversely, if roots partition into two nonempty orthogonal sets, the span of either is a nonzero proper subspace. Reflections belonging to that set preserve its span; reflections belonging to the other set fix it pointwise. This is a proper invariant subspace, contradicting `IsIrreducible`. □

The constructor `RootPairing.ofBilinear` is not a constructor for an arbitrary prescribed finite subset. It takes *all* reflective vectors of its form. For a positive definite real form, every nonzero vector is reflective, because its squared norm is nonzero and divides every real number. In particular, using it directly over $\mathbb{R}$ in positive dimension would give an infinite index set. The direct structure construction in Lemma 9.1 is the interface we actually use.

Lemma 9.2. *Let Φ, Ψ be nonempty Root Systems with bases $(\alpha_i)_{i \in I}$ and $(\beta_i)_{i \in I}$ indexed by the same finite set. Suppose their Cartan matrices coincide and their common nonzero off-diagonal support is connected. The linear map taking each α_i to β_i is a similarity of the ambient Euclidean spaces. If that linear map sends Φ onto Ψ, it is a similarity of the Root Systems themselves.*

Proof. Write $q(i) = \langle \alpha_i, \alpha_i \rangle$ and $r(i) = \langle \beta_i, \beta_i \rangle$. Symmetry of the two inner products gives

$$M(i,j)q(j) = M(j,i)q(i), \qquad M(i,j)r(j) = M(j,i)r(i).$$

On an edge, both matrix entries are nonzero. Dividing shows $r(i)/q(i) = r(j)/q(j)$. Connectedness makes this ratio a single positive constant a. For every pair, including nonedges and equal indices,

$$\langle \beta_i, \beta_j \rangle = \tfrac{1}{2} M(i,j) r(j) = a \langle \alpha_i, \alpha_j \rangle.$$

Expanding arbitrary vectors in the two bases proves the similarity identity. The last assertion follows from the given equality of root sets under the map. Only one common scaling factor is used; short and long roots cannot be rescaled independently. □

Theorem B. [CGSS, Remark 3.11] Every nonempty indecomposable Root System of Definition 2.1 is, after one common positive rescaling, isometric to exactly one of the coordinate Root Systems

$$\mathrm{A}_\ell\ (\ell \geq 1), \quad \mathrm{B}_\ell\ (\ell \geq 2), \quad \mathrm{C}_\ell\ (\ell \geq 3), \quad \mathrm{D}_\ell\ (\ell \geq 4), \quad \mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8, \mathrm{F}_4, \mathrm{G}_2.$$

Exactly the $\mathrm{B}, \mathrm{C}, \mathrm{F}, \mathrm{G}$ systems have two root lengths. The rank-2 model C_2 is similar to B_2; we omit it from the uniqueness list. Every Root System is an orthogonal sum of these indecomposable systems, with an independent scale for each nonzero orthogonal component.

Proof. Apply Lemma 9.1. The theorem `RootPairing.nonempty_base` supplies a base b for this finite, reduced, crystallographic root system over $\mathbb{R}$. Its support is nonempty by `RootPairing.Base.support_nonempty`, since the root index set is nonempty and $2 \neq 0$ in $\mathbb{R}$. The declarations `RootPairing.Base.cartanMatrix_isFiniteCartan` and `RootPairing.Base.cartanMatrix_isIndecomposable` give exactly the hypotheses of Theorem A. Apply it and reindex the base by its resulting bijection.

Section 8 supplies a coordinate Root System with a root base having the resulting matrix. Apply Lemma 9.1 to that model. Use `RootPairing.Base.mk'` with the verified independence and signed expansions to obtain a Mathlib base b_2. Let e be the resulting equivalence between the two base supports and take $f = b$`.toWeightBasis.equiv` b_2`.toWeightBasis` e. This linear equivalence takes each selected simple root to its model simple root, by `Module.Basis.equiv_apply` and `RootPairing.Base.toWeightBasis_apply`. Apply `RootPairing.Base.apply_mem_range_root_of_cartanMatrixEq` to f, and then to f^{-1} with e^{-1} and the Cartan equality reversed. The first application gives $f(\Phi) \subseteq \Psi$ and the second gives $f^{-1}(\Psi) \subseteq \Phi$. Thus $f(\Phi) = \Psi$. Lemma 9.2 makes f an isometry after one common rescaling. Section 12 states all the hypotheses of these declarations; the field, finiteness, spanning, reducedness and crystallographic hypotheses have all been checked above.

For uniqueness we will use invariants of the Root System itself, so no conjugacy theorem for bases is needed. Rank, number of roots, the number of distinct lengths, and the numbers of short and long roots are invariant under similarity. We will prove their values in Lemma 10.2 and record them in Table 2. Among one-length systems, at rank ℓ the classical root counts are $\ell(\ell+1)$ and $2\ell(\ell-1)$; these are equal only at $\ell = 3$, outside our D range. At ranks 6, 7 and 8, the exceptional counts 72, 126 and 240 differ from both classical counts, which are respectively $(42, 60)$, $(56, 84)$ and $(72, 112)$. Among two-length systems, G_2 has squared-length ratio 3; all the others have ratio 2. Type F_4 has 48 roots in rank 4, where $\mathrm{B}_4, \mathrm{C}_4$ have 32. For $\ell \geq 3$, the numbers of short roots in $\mathrm{B}_\ell, \mathrm{C}_\ell$ are 2ℓ and $2\ell(\ell-1)$ and are unequal. Rank separates parameters. These invariants distinguish every entry of the list.

In rank 2 the linear map with matrix $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ takes the four axis roots of B_2 to the four two-coordinate roots of C_2 and the other four roots to its axis roots. Its transpose times itself is $2I$, so it is the required similarity. The displayed coordinate lengths establish the assertion about which systems have two lengths.

Finally, join two roots whenever they have nonzero inner product. Connected components of this finite graph are mutually orthogonal. Each is closed under reflection in one of its own roots. Indeed, if $\langle \alpha, \beta \rangle \neq 0$, then $\langle \alpha, s_\alpha \beta \rangle = -\langle \alpha, \beta \rangle \neq 0$; so the reflected root is in the same component as α; if that inner product is zero, $s_\alpha \beta = \beta$. Thus each component is a Root System in its span. These spans give an orthogonal direct sum since the roots span the ambient space. The classification applies separately to each component, with a separate positive scale. The zero space contributes no component. □

10. Lengths, lines, and the matrix equations

Lemma 10.1. *Let Φ be a Root System. For every pair $\alpha, \beta \in \Phi$ which is neither perpendicular nor proportional, we have*

$$\frac{4\langle \alpha, \beta \rangle^2}{\langle \alpha, \alpha \rangle \langle \beta, \beta \rangle} \in \{1, 2, 3\}.$$

The ratio of the longer length to the shorter length and the acute angle between their lines are respectively $(1, 60°)$, $(\sqrt{2}, 45°)$ or $(\sqrt{3}, 30°)$. For any length $a > 0$ which occurs in Φ, the roots of length a are a Root System in their span, and this subsystem is an orthogonal sum of systems similar to A_ℓ ($\ell \geq 1$), D_ℓ ($\ell \geq 4$), or $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$.

Proof. For such a pair, the two numbers $2\langle \alpha, \beta \rangle / \langle \alpha, \alpha \rangle$ and $2\langle \alpha, \beta \rangle / \langle \beta, \beta \rangle$ are nonzero integers of the same sign. Their product is positive and, by strict Cauchy-Schwarz for a nonproportional pair, is less than 4. The unordered pair of their absolute values is thus $(1, 1)$, $(1, 2)$ or $(1, 3)$. Dividing the two pairings gives the ratio of squared lengths or its reciprocal. Their product is four times the squared cosine of the acute line angle, giving the stated pairs.

The roots of fixed length are finite, nonzero and span their own span; integrality and reducedness are inherited. A reflection in one of those roots maps each of them to a root of the original system and preserves length. Its formula also keeps the vector in their span. This verifies all Root System conditions. Such a subsystem can be decomposable. The classification in Theorem B makes it an orthogonal sum of one-length systems of types A, D, E, because every unequal-length type in that theorem actually has two lengths. □

Lemma 10.2. *The coordinate Root Systems in Section 8 have the root counts, length counts and line counts in Table 2.*

Proof. There are $\ell(\ell+1)$ ordered distinct-coordinate differences in A_ℓ. There are $4\binom{\ell}{2} = 2\ell(\ell-1)$ signed two-coordinate roots in each of $\mathrm{B}_\ell, \mathrm{C}_\ell, \mathrm{D}_\ell$. The first two types have an additional 2ℓ axis roots; these are short in B and long in C.

For E_8 there are $4\binom{8}{2} = 112$ integer roots. Seven half-coordinate signs can be chosen arbitrarily and the eighth is then determined by the product condition, giving $2^7 = 128$ more roots. In E_7, integer roots either use two coordinates among the first six, giving 60, or are $\pm(\mathbf{u}_7 - \mathbf{u}_8)$, giving 2. In a half-coordinate root the last two signs are opposite, and the product of the first six is -1. Choose five of those signs and the last sign freely, giving $2^5 \cdot 2 = 64$. Thus the total is 126. In E_6, an integer root cannot have any nonzero coordinate among 6, 7, 8: the subspace conditions would make all three nonzero, although an integer root has only two nonzero coordinates. Thus there are $4\binom{5}{2} = 40$. For half-coordinate roots, the first five signs determine sign 8 by their product; signs 6 and 7 are both its negative. This gives $2^5 = 32$ and a total of 72.

In F_4 there are 24 two-coordinate long roots, 8 short axis roots and 16 short half-coordinate roots. In G_2 there are 6 short differences and 6 long roots. All these enumerations use disjoint coordinate patterns, so nothing is counted twice. Reducedness and closure under negation give exactly two roots on every root line. Dividing each root count by 2 gives the line count. □

Lemma 10.3. *Let Φ be a nonempty indecomposable Root System in an n-dimensional Euclidean space, and let $\mathcal{L}$ be its N lines. Choose any unit vector $\mathbf{v}_L$ on each line L. Then*

$$\mathcal{F} := \sum_{L \in \mathcal{L}} \mathbf{v}_L \mathbf{v}_L^T = \frac{N}{n} I. \tag{15}$$

Here $\mathbf{v}_L \mathbf{v}_L^T$ means the operator $\mathbf{x} \mapsto \langle \mathbf{v}_L, \mathbf{x} \rangle \mathbf{v}_L$, which is independent of the sign choice. Unit vectors satisfying (15) form a tight frame.

Proof. Every root reflection permutes the lines and is orthogonal, so conjugating the sum by s_α leaves it unchanged. Thus $\mathcal{F}$ commutes with every s_α. The -1-eigenspace of s_α is precisely $\mathbb{R}\alpha$, so $\mathcal{F}\alpha = t_\alpha\alpha$ for some real scalar t_α. Since $\mathcal{F}$ is symmetric,

$$t_\alpha\langle\alpha,\beta\rangle = \langle\mathcal{F}\alpha,\beta\rangle = \langle\alpha,\mathcal{F}\beta\rangle = t_\beta\langle\alpha,\beta\rangle.$$

Thus these scalars agree for nonorthogonal roots. Their nonorthogonality graph is connected, since two unions of components would be an orthogonal decomposition. All scalars have one value t. Roots span, so $\mathcal{F} = tI$. The trace of a unit rank-one operator is 1, as follows by summing its diagonal entries in an orthonormal basis. Taking traces gives $nt = N$, proving (15). □

Lemma 10.4. *Let N lines span an n-dimensional Euclidean space, with $n > 0$. Choose unit vectors on them, let U have these vectors as columns, and write $G = U^TU$. Define*

$$C = 2G - 2I_N. \tag{16}$$

For a real number θ,

$$(C + 2I_N)(C - \theta I_N) = 0 \quad\Longleftrightarrow\quad UU^T = \frac{N}{n}I_n \quad \text{and} \quad \theta = \frac{2N}{n} - 2.$$

Both conditions are independent of the choices of signs.

Proof. Put $t = (\theta + 2)/2$. Expanding the matrix product gives $4(G^2 - tG)$. If $UU^T = tI$, then $G^2 = tG$. Conversely,

$$G^2 - tG = U^T(UU^T - tI)U.$$

The columns of U span, so for any $\mathbf{x}, \mathbf{y}$ in the ambient space there are coefficient vectors $\mathbf{v}, \mathbf{w}$ with $U\mathbf{v} = \mathbf{x}$ and $U\mathbf{w} = \mathbf{y}$. If the displayed matrix vanishes then $\mathbf{x}^T(UU^T - tI)\mathbf{y} = 0$ for all $\mathbf{x}, \mathbf{y}$, so $UU^T = tI$. Its trace is N, so $t = N/n$. Changing column signs replaces U by UD_ϵ for the diagonal sign matrix $D_\epsilon = \operatorname{diag}(\epsilon)$, with $D_\epsilon^2 = I$. It leaves UU^T unchanged and conjugates G and C by D_ϵ. □

Equation (16) is our explicit convention for the undefined matrix in Remark 3.11. It is the direct extension of the convention preceding Theorem 3.7: vectors of squared norm 2 on the lines have Gram matrix $2G = 2I + C$. The source introduces this normalization for any Root System. For two lengths we normalize representatives *line by line* to squared norm 2 to define this matrix; these normalized representatives are not asserted to be a Root System. The entries of C need not be integers: at 45° or 30° they are $\pm\sqrt{2}$ or $\pm\sqrt{3}$. The identities below verify that this convention gives every printed constant of Remark 3.11. Choosing actual roots of unequal lengths instead would give a Gram matrix with nonconstant diagonal, which cannot be $2I + C$ for a zero-diagonal C.

Theorem C. [CGSS, Remark 3.11] With C defined by (16), all the following statements hold. Nonperpendicular, nonproportional roots have the three length-ratio and angle possibilities of Lemma 10.1, and roots of any occurring fixed length form a Root System in their span, which is an orthogonal sum of systems similar to the one-length models A_ℓ ($\ell \geq 1$), D_ℓ ($\ell \geq 4$), $\mathrm{E}_6, \mathrm{E}_7, \mathrm{E}_8$. For $n \geq 2$, the B_n and C_n models give the same n^2 lines in dimension n, with possible distinct-line angles $90°, 60°, 45°$, and

$$(C + 2I)(C - 2(n-1)I) = 0.$$

Type F_4 gives 24 lines in dimension 4 with distinct-line angles $90°, 60°, 45°$, and

$$(C + 2I)(C - 10I) = 0.$$

Type G_2 gives 6 lines in dimension 2 with distinct-line angles $90°, 60°, 30°$, and

$$(C + 2I)(C - 4I) = 0.$$

For $B_2 = C_2$ as line systems, the angle $60°$ does not occur; for $n \geq 3$ all three listed B_n angles occur. A set of three lines pairwise at $45°$ exists in $\mathbb{R}^3$ and is not contained in the lines of any Root System.

Proof. The first assertions are Lemma 10.1. The B_n and C_n axis roots lie on the same axes, and their two-coordinate roots are identical. Thus their lines coincide. Lemma 10.2 gives $n^2, 24, 6$ lines for the three cases. The squared lengths in B, C, F have ratio 2, and those in G have ratio 3. Lemma 10.1, with the perpendicular case included, proves the asserted allowed angle sets. For B_n with $n \geq 3$, the axes 1 and 2 give $90°$, axis 1 and $\mathbf{u}_1 + \mathbf{u}_2$ give $45°$, and $\mathbf{u}_1 + \mathbf{u}_2, \mathbf{u}_1 + \mathbf{u}_3$ give $60°$. The same pairs are present in the C_n line set and in F_4. For $n = 2$ there are only the two axes and the two diagonal lines; their angles are $45°$ or $90°$. In G_2, the two short lines $\mathbf{u}_1 - \mathbf{u}_2, \mathbf{u}_1 - \mathbf{u}_3$ give $60°$; the long root $2\mathbf{u}_1 - \mathbf{u}_2 - \mathbf{u}_3$ is perpendicular to $\mathbf{u}_2 - \mathbf{u}_3$ and makes a $30°$ line angle with $\mathbf{u}_1 - \mathbf{u}_2$, since its inner product with that root is 3 and the product of their lengths is $\sqrt{12}$.

The systems are indecomposable by Lemma 8.7. Apply Lemmas 10.3 and 10.4. The values of N/n are $n, 6, 3$; thus $\theta = 2n - 2, 10, 4$. We will also give explicit coordinate sums verifying the frame identities in Appendix A.

For the final example put $c = 1/\sqrt{2}$ and use the three unit vectors

$$\mathbf{u}_1, \qquad c\mathbf{u}_1 + c\mathbf{u}_2, \qquad c\mathbf{u}_1 + (1-c)\mathbf{u}_2 + \sqrt{2c-1}\mathbf{u}_3.$$

The radicand is positive. The squared norm of the third vector is $c^2 + (1-c)^2 + 2c - 1 = 2c^2 = 1$. Their pairwise inner products are $c, c, c^2 + c(1-c) = c$, so the line angles are all $45°$. If roots on these lines belonged to a Root System, every pair of their squared lengths would have ratio 2 in one order. Choose one with minimal squared length b. Each of the other two must then have squared length $2b$, but their equal lengths contradict their $45°$ angle by Lemma 10.1. This proves noncontainment, in any ambient dimension and without an irreducibility assumption. □

The same reasoning proves the equations of Theorem 3.7, independently of its trace calculation. The values of N/n for A_n, D_n, E_6, E_7, E_8 are $(n+1)/2, n-1, 6, 9, 15$, respectively. Thus their second constants are $n - 1, 2(n-2), 10, 16, 28$. In particular the printed 18 for E_8 is false; 28 is required. In fact the squared-norm-2 Gram matrix $2I + C$ has nonzero eigenvalue 30 under the frame identity, making the corresponding C eigenvalue 28.

11. Interpretations and corrections of the sources

We read [CGSS] in its published form, in particular pages 312–315. The following points are part of our statements.

(1) The finite-type matrix assertion requires a nonempty index set. The empty matrix satisfies both Mathlib predicates and is a separate rank-zero case. For Root Systems we include nonemptiness in "indecomposable", matching the nontriviality in `RootPairing.IsIrreducible`.

(2) We use nonzero roots in Definition 3.9 and read the S in its condition (b) as Σ. Zero is excluded by the definition's own proportionality condition, as explained in Section 2.

(3) The opening of Remark 3.11 needs "not proportional" as well as "not perpendicular". Taking $\beta = \alpha$ gives product 4 and line angle $0°$. Also, a length ratio is understood as longer divided by shorter; an ordered ratio can instead be a reciprocal.

(4) A fixed-length subset is a Root System in *its own span*. It need not span the original ambient space, and it can be decomposable. For example the long roots of B_2 are two perpendicular sign pairs, so their subsystem is $A_1 \oplus A_1$. For a failure of spanning, $\{\pm\mathbf{u}_1, \pm 2\mathbf{u}_2\}$ is a Root System in $\mathbb{R}^2$, but its roots of length 1 span only one axis. Lemma 10.1 and Theorem C also state that these subsystems are orthogonal sums of the A, D, E models.

We prove this part of Remark 3.11 from Theorem B, without using [CGSS, Theorem 3.10]; Theorem B does not depend on the assertion.

(5) We define the matrix in Remark 3.11 by $C = 2G - 2I$ for unit representatives, extending the equal-length convention before Theorem 3.7. The source supplies no separate definition there. The resulting three equations agree with the printed ones. This is an explicit convention, not a claim that the unnormalized Gram matrix of unequal-length roots is $2I + C$.

(6) The angle lists mean allowed angles. At rank 2, the $\mathrm{B}_n, \mathrm{C}_n$ list has no $60°$ pair. At rank 1 there is one line, no pair of distinct lines and only one root length; its matrix is $C = [0]$. The unequal-length statement starts at rank 2.

(7) The constant 18 for E_8 in Theorem 3.7 (p. 312) is an error of the paper. We correct it to 28.

(8) Theorem 3.8 is for finite connected simple graphs. We draw the graphs of its Figure 1 in our Figure 1. The printed labels A_n, D_n, E_6, E_7, E_8 correspond respectively to our $\mathrm{Cyc}_{n+1}, \widetilde{\mathrm{D}}_n, \widetilde{\mathrm{E}}_6, \widetilde{\mathrm{E}}_7, \widetilde{\mathrm{E}}_8$. Corollary D strengthens the source's one-way classification to an equivalence. The asterisk in each source drawing marks a vertex whose deletion leaves the finite diagram with the same type label; we omit the asterisks because we prove the basis assertion for every deletion, so also for those marked vertices. The deletion assertion holds for *any* one vector, in the span of dimension $m - 1$. Its notation $\mathbb{R}^n$ must refer to that span (or to a spanning realization), rather than an arbitrarily larger ambient space.

Theorem 3.12 is outside the scope of this document; we prove it in `cgss/proofs/thm312/thm312.tex`. The three-line example immediately before it is proved here. No classification theorem from [CGSS] or [BCN] is used as a premise. In the Mathlib interface we additionally distinguish the actual rank of `CartanMatrix.A` from its docstring, and the all-reflective-vectors constructor from a constructor for a prescribed finite root set.

12. Interface with Mathlib

We will distinguish library premises, available results that we do not use as premises, and the missing statements proved here. All paths below are relative to Mathlib at the commit stated in the introduction, the version pinned in `lake-manifest.json`, and we read the declarations there. We construct the coordinate models here; we do not cite them as existing Mathlib declarations.

12.1. **Matrices and Rayleigh quotients.** The file `Mathlib/LinearAlgebra/Matrix/Cartan.lean` defines `CartanMatrix.A`, `CartanMatrix.B`, `CartanMatrix.C`, `CartanMatrix.D`, and `CartanMatrix.E`, each with argument $n : \mathbb{N}$ and value a matrix on `Fin n` with entries in $\mathbb{Z}$. It also defines `CartanMatrix.F`$_4$ and `CartanMatrix.G`$_2$, on `Fin 4` and `Fin 2`. The identifiers `CartanMatrix.E`$_6$, `CartanMatrix.E`$_7$, and `CartanMatrix.E`$_8$ are deprecated aliases for `CartanMatrix.E 6`, `CartanMatrix.E 7`, and `CartanMatrix.E 8`. They are not separate conventions. The file's small-rank declarations include `CartanMatrix.B_one`, `CartanMatrix.C_one`, `CartanMatrix.D_one`, `CartanMatrix.D_two`, `CartanMatrix.B_two`, `CartanMatrix.C_two`, `CartanMatrix.D_three`, and `CartanMatrix.D_three'`. They have no hypotheses, and give the identities stated in Remark 6.1. We checked those identities directly as well. The same file contains `CartanMatrix.E_det`: for $n : \mathbb{N}$ and $3 \le n$, it states $\det(\mathrm{E}_n) = 9 - n$, with the subtraction in $\mathbb{Z}$. We also prove this determinant formula for the ranks used here by the leaf expansion in Appendix A.

`Matrix.IsFiniteCartan`, in that file, takes a type ι with `[Fintype iota]` and `[DecidableEq iota]`, and an integer square matrix M. Its four fields are `diag`, `offDiag_nonpos`, `zero_comm`, and `exists_posDef`, with exactly the conditions in Section 2. The last is

$$\exists d : \iota \to \mathbb{Z}, \quad (\forall i,\ 0 < d(i)) \ \wedge\ (\mathrm{diag}(d)M).\texttt{PosDef}.$$

The definition `Matrix.PosDef`, in `Mathlib/LinearAlgebra/Matrix/PosDef.lean`, uses Hermitian symmetry and positivity on nonzero finitely supported vectors over the same coefficient ring. Its specialization here is over $\mathbb{Z}$, not $\mathbb{R}$. Lemma 3.1 proves the needed change of scalars. In the same file, `Matrix.posDef_iff_dotProduct_mulVec` assumes `[Fintype n]`, `[Ring R]`, `[PartialOrder R]` and `[StarRing R]`. For H : `Matrix n n R` it states

$$H.\texttt{PosDef} \iff H.\texttt{IsHermitian} \wedge \forall \mathbf{x} \neq 0, \quad 0 < \operatorname{star}(\mathbf{x}) \cdot (H\mathbf{x}).$$

Over $\mathbb{Z}$ or $\mathbb{R}$ this is exactly our quadratic-form condition.
The file `Mathlib/LinearAlgebra/Matrix/ToLinearEquiv.lean` supplies `Matrix.exists_mulVec_eq_zero_iff`. For `[Fintype n]`, `[DecidableEq n]`, `[CommRing R]`, `[IsDomain R]` and H : `Matrix n n R`, its conclusion is $(\exists \mathbf{v} \neq 0,\ H\mathbf{v} = 0) \leftrightarrow \det H = 0$. We apply it over $\mathbb{R}$ and $\mathbb{Z}$ in Lemma 3.1. The declaration `RingHom.map_det`, in `Mathlib/LinearAlgebra/Matrix/Determinant/Basic.lean`, takes `[Fintype n]`, `[DecidableEq n]`, commutative rings R, S, a ring homomorphism $f : R \to S$ and a square matrix H over R. It states $f(\det H) = \det(f(H))$, with f applied entrywise.
`Matrix.IsIndecomposable`, in `Mathlib/LinearAlgebra/Matrix/Block.lean`, requires only a type of indices and `[Zero R]` for the entries. It says that for every i, j : $\mathbb{N}$, blocks A, B, D and equivalence e : `Fin i` $\oplus$ `Fin j` $\simeq \iota$, the equality $M =$ `Matrix.reindex` $e\ e \begin{pmatrix} A & B \\ 0 & D \end{pmatrix}$ implies $i = 0$ or $j = 0$. There is no nonempty or finite typeclass hypothesis in this definition. Its equivalence with connected support for the matrices at issue is proved in Lemma 3.2.
The file `Mathlib/Analysis/InnerProductSpace/Rayleigh.lean` supplies the following exact analytic interfaces. The shared hypotheses are a scalar type $\mathbb{k}$ with `[RCLike k]`, a type V with `[NormedAddCommGroup V]` and `[InnerProductSpace k V]`.

- `LinearMap.IsSymmetric.hasEigenvalue_iSup_of_finiteDimensional` additionally has `[FiniteDimensional k V]`, `[Nontrivial V]`, a linear endomorphism ϕ, and h_ϕ : ϕ.`IsSymmetric`. It asserts that the real supremum $\sup_{\mathbf{x}\neq 0} \operatorname{Re}\langle \phi\mathbf{x}, \mathbf{x}\rangle / \|\mathbf{x}\|^2$, cast to $\mathbb{k}$, is an eigenvalue of ϕ.
- `IsSelfAdjoint.hasEigenvector_of_isMaxOn` additionally has `[CompleteSpace V]`, a continuous linear endomorphism ϕ, h_ϕ : `IsSelfAdjoint phi`, a vector $\mathbf{x}_0 \neq 0$, and `IsMaxOn phi.reApplyInnerSelf (sphere 0 (norm x0)) x0`. It makes $\mathbf{x}_0$ an eigenvector for the same supremum over nonzero vectors. The function maximized in this hypothesis is the real part of $\langle \phi\mathbf{x}, \mathbf{x}\rangle$ on that sphere.

For our application $\mathbb{k} = \mathbb{R}$ and $V = \mathbb{R}^I$ with its Euclidean inner product, where I is finite and nonempty. The abbreviation `Matrix.toEuclideanLin`, in `Mathlib/Analysis/InnerProductSpace/PiL2.lean`, takes `[RCLike k]`, `[Fintype n]` and `[DecidableEq n]` and sends a matrix on row type m and column type n to the linear map between the corresponding Euclidean spaces. In `Mathlib/Analysis/Matrix/Hermitian.lean`, the declaration `Matrix.isSymmetric_toEuclideanLin_iff` assumes `[RCLike k]`, `[Fintype n]`, `[DecidableEq n]` and a square matrix B; it states that B.`toEuclideanLin` is symmetric if and only if B is Hermitian. This is the current name for the matrix bridge; `Matrix.isHermitian_iff_isSymmetric` in the same file is its deprecated reversed equivalence, with the same hypotheses. Over $\mathbb{R}$, symmetry of B supplies the Rayleigh hypothesis. Normalizing any nonzero vector preserves its quotient, so the supremum over nonzero vectors in the library equals our maximum on the unit sphere. We included an elementary proof of these extremum facts in Lemma 4.1; the positivity, simplicity and strict monotonicity consequences follow from our proofs, without a Perron-Frobenius declaration.

12.2. **The root-system interface actually used.** For precision, a *common root-pairing context* in the following list means types ι, R, M, N with

$$\texttt{[CommRing R]},\quad \texttt{[AddCommGroup M]},\quad \texttt{[Module R M]},$$
$$\texttt{[AddCommGroup N]},\quad \texttt{[Module R N]},\qquad P : \texttt{RootPairing}\ \iota\ R\ M\ N.$$

When `[Field R]` is stated it supplies the commutative ring instance. The symbol M in this list is a module type, not the Cartan matrix. Finiteness below refers to the entire root index type ι, not just the support of a base. We state explicitly whether the source asks for `Finite` or `Fintype`.

(1) In `Mathlib/LinearAlgebra/RootSystem/Defs.lean`, `RootPairing` is the structure of a perfect bilinear pairing, embeddings of roots and coroots, their self-pairing 2, and the two reflection identities. It is defined in the common context without finiteness or spanning assumptions. Its `pairing i j` is the evaluation of root i on coroot j, exactly (2) in our model. `RootPairing.IsRootSystem`, with no additional hypotheses, demands that both the roots and the coroots span their modules. For our perfect pairing, $\texttt{inner}_{\mathbb{I}}$ in `Mathlib/Analysis/InnerProductSpace/Basic.lean` assumes `[SeminormedAddCommGroup V]` and `[InnerProductSpace Real V]` and is the real bilinear inner product on V. The declaration `LinearMap.IsPerfPair.of_injective`, in `Mathlib/LinearAlgebra/PerfectPairing/Basic.lean`, assumes `[Field K]`, additive commutative groups M, N with K-module structures, `[FiniteDimensional K M]`, and a bilinear map $p : M \to$ `Module.Dual K N`. Injectivity of p and of its flip gives p`.IsPerfPair`. In Lemma 9.1 both modules are the finite-dimensional normed space V over $\mathbb{R}$, and definiteness proves both injections.

(2) In `Mathlib/LinearAlgebra/RootSystem/IsValuedIn.lean`, `RootPairing.IsCrystallographic` abbreviates `P.IsValuedIn Int`. In the common context it demands $\forall i, j,\ \exists z \in \mathbb{Z}$, the image of z in R equals P.`pairing`(i, j). There is no order assumption on R. In `Mathlib/LinearAlgebra/RootSystem/Reduced.lean`, `RootPairing.IsReduced` demands that for every i, j, failure of linear independence of the pair of roots implies equality or equality to the negative. It also uses only the common context.

(3) In `Mathlib/LinearAlgebra/RootSystem/Irreducible.lean`, `RootPairing.IsIrreducible` uses the common context and asserts nontriviality of both modules and that any nonzero submodule invariant under every root reflection, or every coroot reflection respectively, is the whole module. The constructor `RootPairing.IsIrreducible.mk'` takes a field K, the two K-module structures and additive commutative group structures, `[Nontrivial M]`, a root pairing over K, and the hypothesis that every nonzero root-reflection-invariant submodule of M is all of M. It concludes `P.IsIrreducible`; it does not require finiteness, characteristic zero, reducedness or crystallographic pairings. Lemma 9.1 proves both halves directly and explains the meaning of this constructor.

(4) In `Mathlib/LinearAlgebra/RootSystem/Base.lean`, `RootPairing.Base` is defined in the common context. Its support is a finite set of root indices, with linear independence on roots and coroots, and signed membership in the additive monoids generated by the corresponding simple roots and simple coroots. `RootPairing.Base.toWeightBasis` takes a base b and `[P.IsRootSystem]`, with no other extra hypotheses, and returns a module basis of M indexed by b`.support`. Its value at i is P`.root`(i), also stated as `RootPairing.Base.toWeightBasis_apply`. The declaration `RootPairing.Base.support_nonempty`, in the same file, takes a base in the common context with `[Nonempty iota]` and `[NeZero (2 : R)]` and concludes b`.support.Nonempty`. It does not require finiteness of all roots or `IsRootSystem`.

(5) In `Mathlib/LinearAlgebra/RootSystem/BaseExists.lean`, both `RootPairing.nonempty_base` and `RootPairing.Base.mk'` have `[Finite iota]`, `[Field R]`, `[CharZero R]`, the two module and additive commutative group structures, and `[P.IsRootSystem]`, `[P.IsCrystallographic]`, `[P.IsReduced]`. The first asserts `Nonempty P.Base`; it does not assume `Nonempty iota`. The second takes a subset s of root indices, linear independence of P.`root` on s, and for every root the assertion that it or its negative belongs to the additive monoid generated by P.`root`(s). It returns P.`Base`. In a finite set of generators this monoid consists precisely of finite nonnegative integer sums: closure contains such sums by repeated addition, and the set of such sums is itself an additive submonoid containing the generators. Thus our expansion checks give exactly the required hypothesis. The library bridge is `Submodule.span_nat_eq_addSubmonoidClosure`, in `Mathlib/LinearAlgebra/Span/Defs.lean`: for `[AddCommMonoid M]` and s : `Set M`, it identifies the underlying additive submonoid of `Submodule.span` $\mathbb{N}$ s with `AddSubmonoid.closure` s. It needs no finiteness hypothesis on s.

(6) In `Mathlib/LinearAlgebra/RootSystem/CartanMatrix.lean`, `RootPairing.Base.cartanMatrix` needs the common context, a base b and `[P.IsCrystallographic]`. It is the integer matrix on b.`support` with entries P.`pairingIn` $\mathbb{Z}$ i j. `RootPairing.Base.cartanMatrix_isFiniteCartan` has, in addition, `[CharZero R]`, `[IsDomain R]`, `[Finite iota]`, `[DecidableEq iota]`, and `[P.IsRootSystem]`. It asserts that b.`cartanMatrix` satisfies `Matrix.IsFiniteCartan`. It does not need reducedness or irreducibility as additional hypotheses.

(7) In that same file, `RootPairing.Base.cartanMatrix_isIndecomposable` has the common context, a base b, `[P.IsCrystallographic]`, `[CharZero R]`, `[IsDomain R]`, `[Finite iota]`, `[P.IsReduced]`, and `[P.IsIrreducible]`. It concludes b.`cartanMatrix.IsIndecomposable`. Its underlying declaration `RootPairing.Base.induction_on_cartanMatrix` has the same hypotheses. It takes a predicate p on the support, indices i, j, $p(i)$, and

$$\forall u, v, \quad p(u) \longrightarrow b.\texttt{cartanMatrix}(v, u) \neq 0 \longrightarrow p(v),$$

and concludes $p(j)$. Neither declaration adds `[DecidableEq iota]` or `[P.IsRootSystem]` explicitly. We use the indecomposability declaration; the induction declaration explains its connectedness content.

(8) Also in `Mathlib/LinearAlgebra/RootSystem/CartanMatrix.lean`, `RootPairing.Base.apply_mem_range_root_of_cartanMatrixEq` has `[CommRing R]`, `[CharZero R]`, `[IsDomain R]`, two root pairings P, P_2 over R with their additive commutative root and coroot groups and R-module structures, `[Finite iota]`, and `[P.IsRootSystem]`, `[P.IsCrystallographic]`, `[P.IsReduced]`, `[P2.IsCrystallographic]`. Its data include bases b, b_2, an equivalence $e : b.\texttt{support} \simeq b_2.\texttt{support}$, and the hypothesis

$$\forall i, j, \quad b_2.\texttt{cartanMatrix}(e(i), e(j)) = b.\texttt{cartanMatrix}(i, j).$$

It also takes a linear equivalence $f : M \simeq_R M_2$ satisfying $f(P.\texttt{root}(i)) = P_2.\texttt{root}(e(i))$ for every simple-root index i, and $m \in \text{range}(P.\texttt{root})$. It concludes $f(m) \in \text{range}(P_2.\texttt{root})$. This direction requires no finiteness, spanning or reducedness hypothesis on the target pairing. We have those hypotheses for both pairings, so we apply it again with source and target interchanged to obtain equality of root sets.

In `Mathlib/LinearAlgebra/Basis/Defs.lean`, `Module.Basis.equiv` and `Module.Basis.equiv_apply` take `[Semiring R]`, additive commutative monoids M, M_2 with R-module structures, bases b, b_2 and an equivalence e of their index types. They give a linear equivalence with b.`equiv` b_2 e $(b(i)) = b_2(e(i))$, without finiteness assumptions. Applying them to the weight bases constructs the f used in Theorem B.

For comparison, `RootPairing.Base.equivOfCartanMatrixEq` in the Cartan-matrix file has the same root-pairing hypotheses and additionally `[Finite iota2]`, `[P2.IsRootSystem]`, `[P2.IsReduced]`. From b, b_2, e and the displayed Cartan equality it constructs P.`Equiv` P_2, without an irreducibility hypothesis. Theorem B uses the root-preservation lemma in both directions, so it does not depend on unfolding this constructor.

Every additional hypothesis in the uses of these declarations in Theorem B is supplied by $R = \mathbb{R}$, finiteness of the given root set, Lemma 9.1, or the coordinate verifications. Decidable equality on a finite index set can be chosen classically; this is a logical choice, not an unproved mathematical assumption.

12.3. **Other Mathlib files and what they provide.** These results are available but are not additional classification premises in our proof.

- `Mathlib/LinearAlgebra/RootSystem/OfBilinear.lean` defines `LinearMap.IsReflective B x` in the commutative-ring module context. Its fields are regularity of $B(\mathbf{x}, \mathbf{x})$ and $B(\mathbf{x}, \mathbf{x}) \mid 2B(\mathbf{x}, \mathbf{y})$ for every $\mathbf{y}$. The constructor `RootPairing.ofBilinear` assumes `[Module.IsReflexive R M]`, a bilinear form B, its `LinearMap.Nondegenerate` and `LinearMap.IsSymm` properties, and `IsRegular (2 : R)`. Its root index set is $\{\mathbf{x} : M \mid$ `LinearMap.IsReflective B x`$\}$, and its coroot space is `Module.Dual R M`. It has no prescribed-root-subset parameter. This is why it is not applied directly in our proof.
- `Mathlib/LinearAlgebra/RootSystem/Finite/Lemmas.lean` contains `RootPairing.InvariantForm.exists_apply_eq_or`. Its hypotheses are the common root-pairing context with `[Finite iota]`, `[CharZero R]`, `[IsDomain R]`, `[P.IsCrystallographic]`, `[P.IsReduced]`, `[P.IsIrreducible]`, an invariant form $B : P$.`InvariantForm`, and `[Nonempty iota]`. Its conclusion is
$$\exists i, j, \ \forall k, \quad B(\alpha_k, \alpha_k) = B(\alpha_i, \alpha_i) \text{ or } B(\alpha_k, \alpha_k) = B(\alpha_j, \alpha_j).$$
An invariant form, as defined in `Mathlib/LinearAlgebra/RootSystem/RootPositive.lean`, is a symmetric bilinear form, nonzero on each root's self-pairing, for which all root reflections are orthogonal. It need not be a positive definite form. Our Euclidean inner product meets these requirements.

  In the same finite-lemmas file, `RootPairing.forall_pairing_eq_swap_or` has the same ring, module, finiteness, crystallographic, reduced and irreducible hypotheses, but no invariant-form argument and no nonempty-index hypothesis. It asserts the disjunction of the following two global statements: for all i, j, either $p(i, j) = p(j, i)$, or one is twice the other; for all i, j, either $p(i, j) = p(j, i)$, or one is three times the other. Thus the choice of 2 or 3 is global, not a choice made separately for each pair. These results supply the two-length bound, but not the list of finite-type Cartan matrices. We derive the lengths directly from our models and prove the pairwise angle statement independently.
- `Mathlib/LinearAlgebra/RootSystem/Finite/Nondegenerate.lean` proves positivity and nondegeneracy used internally by the Cartan theorems. For example `RootPairing.rootForm_nondegenerate` assumes the common context, `[Fintype iota]`, `[IsDomain R]`, `[P.IsAnisotropic]` and `[P.IsRootSystem]`, and concludes P.`RootForm.Nondegenerate`. `RootPairing.instIsAnisotropicOfIsCrystallographic`, in the same file, supplies anisotropy from the common context, `[Fintype iota]`, `[CharZero R]` and `[P.IsCrystallographic]`. We do not identify this canonical form with the given Euclidean metric. Our similarity argument explicitly recovers the latter from the simple-root lengths.

- `Mathlib/LinearAlgebra/RootSystem/Finite/G2.lean` studies an embedded long-short pair. The class `RootPairing.EmbeddedG2`, in the common context, extends crystallographic and reduced and supplies indices `long`, `short` with integer pairing -3 in that order. With `[Finite iota]`, `[CharZero R]`, `[IsDomain R]`, `[P.EmbeddedG2]` and `[P.IsIrreducible]`, `RootPairing.EmbeddedG2.mem_allRoots` puts every root in its displayed list of twelve, and `RootPairing.EmbeddedG2.card_index_eq_twelve` concludes `Nat.card` $\iota = 12$. Under those same hypotheses, `RootPairing.EmbeddedG2.basis` is a module basis on `Fin 2`, in the order short, long. It is not bundled as a `RootPairing.Base`; the file has an explicit TODO to strengthen it. Our base is independently checked, in the order long, short, so its Cartan matrix is precisely `CartanMatrix.`G_2.
- `Mathlib/NumberTheory/ADEInequality.lean` contains `ADEInequality.classification`, with arguments $p, q, r : \mathbb{N}_{>0}$ and no other hypotheses. It states $1 <$ `ADEInequality.sumInv`$\{p, q, r\}$ if and only if `ADEInequality.Admissible`$\{p, q, r\}$. The rational sum is $p^{-1} + q^{-1} + r^{-1}$, and admissibility means an unordered triple $\{1, q', r'\}$, $\{2, 2, r'\}$, or one of $\{2, 3, 3\}, \{2, 3, 4\}, \{2, 3, 5\}$, with positive integer parameters. This is the numerical counterpart of the three-arm classification. Lemma 5.5 instead gives the actual obstructing subgraphs in every excluded case; the inequality theorem alone does not supply those graph embeddings or handle multiple edges.

12.4. **What this document adds.** The new statements needed on top of these interfaces are the integer-to-real positivity lemma; equivalence of connected support and the matrix predicate; the symmetric nonnegative-matrix positivity, simplicity and strict monotonicity lemmas; the affine eigenvector data and the full weighted dichotomy; orientation, nonempty finite-type classification and uniqueness; the coordinate Root Systems with bases and their exact index match; the Euclidean finite-root-set constructor and the equivalence of the two irreducibility notions; the passage from equality of Cartan matrices to a single similarity; and the line-frame identities, counts and Smith deletion assertion. We proved these statements in the preceding sections, including the explicit path and three-arm labellings in Lemmas 5.3 and 5.4. We do not cite the coordinate models or the finite-type matrix classification as existing Mathlib declarations.

13. Lean statement shapes

We will retain the settled names in `cgss/notation.md`, Section 10. The following are the statement shapes, without proof bodies, that we wrote before the formalization. The Lean formalization, in `LineGraphs/Cartan/` and `LineGraphs/RootSetClassification/`, follows them in substance; where its names differ, `cgss/notation.md`, Section 10, gives the final ones. The auxiliary names introduced here are not Mathlib declarations. We state their mathematical meanings explicitly.

For Theorem A, let the proposed predicate `CartanMatrix.IsFiniteTypeModel N`, for an integer matrix N on `Fin l`, mean the disjunction of the equalities to the matrices in (1), with exactly the rank restrictions in that list. In particular, its B clause is `And (2 <= l) (N = CartanMatrix.B l)`; its C clause starts at rank 3, and its F_4 and G_2 clauses include the corresponding rank equalities and transport of indices. The existence part of the settled classification name has shape

```
theorem Matrix.IsFiniteCartan.classification
    {I : Type*} [Fintype I] [DecidableEq I] [Nonempty I]
    {M : Matrix I I Int}
    (hM : M.IsFiniteCartan) (hI : M.IsIndecomposable) :
    Exists (fun l : Nat => Exists (fun e : Equiv (Fin l) I =>
      CartanMatrix.IsFiniteTypeModel (M.submatrix e e)))
```

Thus a B_ℓ outcome is precisely an equivalence `e : Equiv (Fin l) I` with `M.submatrix e e = CartanMatrix.B l`. The auxiliary theorem `CartanMatrix.IsFiniteTypeModel.type_eq_of_reindex_eq` will state that two permitted type-and-rank choices whose matrices are equal after reindexing have the same type and rank. The invariants in Section 6 prove this uniqueness assertion. We do not assert uniqueness of e. For the recognition step, the name `SimpleGraph.nonempty_iso_pathGraph` (rule 1 of `cgss/notation.md` §10 asks for the `nonempty_iso` prefix) has the following shape:

```
theorem SimpleGraph.nonempty_iso_pathGraph
    {U : Type*} [Fintype U] [DecidableEq U]
    (X : SimpleGraph U) [DecidableRel X.Adj]
    (hc : X.Connected) (ha : X.IsAcyclic)
    (hd : forall u, X.degree u <= 2) :
    Exists (fun phi : Equiv (Fin (Fintype.card U)) U =>
      forall i j, X.Adj (phi i) (phi j) <->
        Or (i.val + 1 = j.val) (j.val + 1 = i.val))
```

The endpoint version, `SimpleGraph.exists_path_labelling_from`, adds a vertex w with degree at most 1 and concludes that the same bijection sends the first index to w. Connectedness supplies the nonempty vertex type, so this first index exists. For `SimpleGraph.exists_three_arm_labelling`, the hypotheses are finiteness, connectedness, acyclicity, a specified vertex v of degree 3, and degree at most 2 at every other vertex. Its conclusion is the existence of positive a, b, c and an equivalence from `Unit` $\oplus$ (`Fin a` $\oplus$ (`Fin b` $\oplus$ `Fin c`)) to the vertex type, sending the `Unit` element to v, with the edge equivalence of Lemma 5.4. Position t of that lemma is position $t-1$ in its `Fin` arm. These conclusions retain the maps used in Theorem 5.6 and Section 6.

The library definitions in the recognition shape are `SimpleGraph.Connected` in `Mathlib/Combinatorics/SimpleGraph/Connectivity/Connected.lean`, which includes nonemptiness and reachability between every two vertices; `SimpleGraph.IsAcyclic` in `Mathlib/Combinatorics/SimpleGraph/Acyclic.lean`, which excludes cycles in all closed walks; and `SimpleGraph.degree` in `Mathlib/Combinatorics/SimpleGraph/Finite.lean`, which counts the neighbours when the neighbour set has a `Fintype` instance. Our finite vertex type and decidable adjacency supply that instance.

The proposed definition `Matrix.maxRayleigh` will be the maximum in Section 2, for a real symmetric matrix on a finite nonempty index type. Corollary D retains the settled name `SimpleGraph.smith`, with hypothesis `Matrix.maxRayleigh` $(X$`.adjMatrix` $\mathbb{R}) = 2$. Here `SimpleGraph.adjMatrix`, in `Mathlib/Combinatorics/SimpleGraph/AdjMatrix.lean`, assumes decidable adjacency and coefficients with `Zero` and `One`; it has entry 1 on adjacent pairs and entry 0 elsewhere. The conclusion of `SimpleGraph.smith` is a graph isomorphism to one of the settled models `SimpleGraph.smithA l`, `SimpleGraph.smithD l`, `SimpleGraph.smithE`$_6$, `SimpleGraph.smithE`$_7$, or `SimpleGraph.smithE`$_8$; the first is the cycle on $\ell+1$ vertices with $\ell \geq 2$, and the second has $\ell \geq 4$. These model names come from the project notation file, not from Mathlib. Theorem B retains `IsRootSet.classification`. Its indecomposable case concludes the existence of a linear equivalence f onto exactly one coordinate model and $a > 0$ with $\langle f\mathbf{x}, f\mathbf{y}\rangle = a\langle \mathbf{x}, \mathbf{y}\rangle$ and $f(\Phi) = \Psi$. The general case permits an independent scale on each orthogonal component. These shapes use the ordinary proofs above.

14. Further directions

A natural question is how much of the weighted obstruction argument can serve other spectral classifications. The explicit labellings isolate its graph-theoretic content, while the Rayleigh

comparison works for arbitrary nonnegative weights. One may ask for a reusable Mathlib interface that connects these two parts.

Theorem A also suggests a library contribution: a classification with its nonempty hypothesis, uniqueness invariants and coordinate realization maps. A further question is how best to connect that interface with the existing two-length theorem and embedded G_2 infrastructure. The ordinary proofs and rational certificates here supply the mathematical input for such a contribution, as they did for the structured proofs and the Lean formalization of this project.

Appendix A. Finite data and independent checks

We will collect the numerical data so that a formalization can replace each finite check by a kernel proof on integers, rationals, or the indicated square roots. The formulas for arbitrary rank, rather than their checked initial instances, are part of the proofs above.

A.1. **Affine eigenvectors.** Table 1 gives every obstruction used in the dichotomy. For a three-arm tree the arm lists are read outwards from the centre. For $\widetilde{D}_\ell$, every vertex on the path between forks has value 2, including both forks.

Table 1: Positive eigenvectors with eigenvalue 2.

Graph	Entries
Cyc_m, $m \geq 3$	1 at every vertex.
$\widetilde{D}_4$	2 at the centre; 1 at each of its four leaves.
$\widetilde{D}_\ell$, $\ell \geq 5$	2 along the path of $\ell - 4$ edges; 1 at its four attached leaves.
$T(2,2,2)$	Centre 3; each arm $(2,1)$.
$T(1,3,3)$	Centre 4; arms (2), $(3,2,1)$, $(3,2,1)$.
$T(1,2,5)$	Centre 6; arms (3), $(4,2)$, $(5,4,3,2,1)$.
X_m^{end}, $m \geq 3$	In path order: $1, \sqrt{2}, \ldots, \sqrt{2}, 1$.
X_k^{fork}, $k \geq 1$	At $v_0, \ldots, v_{k-1}$: 2; at v_k: $\sqrt{2}$; at the two extra leaves: 1.
X^{mid}	In path order: $1, 2, 3, 2\sqrt{2}, \sqrt{2}$.
X^{triple}	In path order: $\sqrt{3}, 2, 1$.

We proved the local equations in Lemma 5.2; equation (4), with an additional edge-weight factor in each summand, also proves that each affine matrix $2I - B$ has one-dimensional kernel. This follows because every displayed weight and entry is positive.

A.2. **Determinants and bases.** We will record the leading determinants of each named matrix and give explicit rational positivity certificates for the exceptional matrices. Let Δ_k be the determinant of its leading $k \times k$ submatrix and set $\Delta_0 = 1$. The complete lists are

$$
\begin{array}{c|l}
A_\ell & \Delta_k = k+1 \quad (1 \leq k \leq \ell) \\
B_\ell, C_\ell & \Delta_k = k+1 \quad (k < \ell), \quad \Delta_\ell = 2 \\
D_\ell & \Delta_k = k+1 \quad (k < \ell), \quad \Delta_\ell = 4 \\
E_6 & (2,4,6,5,4,3) \\
E_7 & (2,4,6,5,4,3,2) \\
E_8 & (2,4,6,5,4,3,2,1) \\
F_4 & (2,3,2,1) \\
G_2 & (2,1).
\end{array} \tag{17}
$$

We will justify these formulas without assuming a determinant table. We will use a general leaf expansion. Suppose $v \neq w$, the off-diagonal nonzero entries in both row and column v of M occur only at w, and $M(v,v) = 2$. Put $p = M(v,w)M(w,v)$ and use subscripts $-v$ and $-v,-w$ for deletion of those rows and columns. Then

$$\det M = 2\det M_{-v} - p\det M_{-v,-w}. \tag{18}$$

In the determinant expansion a nonzero permutation term either fixes v, contributing to the first summand, or sends v to w. In the second case only row w can supply column v, so the permutation interchanges v, w. Its transposition contributes the minus sign and the factor p, proving the formula without a tridiagonal hypothesis. For a path whose last edge has product p, this specializes to $\Delta_k = 2\Delta_{k-1} - p\Delta_{k-2}$. For a unit path this recurrence, with $\Delta_0 = 1, \Delta_1 = 2$, gives $k+1$ by induction. The last edge product in B, C is 2, giving $2\ell - 2(\ell-1) = 2$. For D_ℓ, the last leaf is $v = \ell - 1$ and its neighbour is $w = \ell - 3$. Deleting that leaf leaves a unit path on $\ell - 1$ vertices, of determinant ℓ. Deleting that leaf and its neighbour leaves a path on $\ell - 3$ vertices and one isolated vertex, with determinant $2(\ell-2)$. The same leaf expansion gives $2\ell - 2(\ell-2) = 4$. For E, the first three determinants are $2, 4, 6$, since the first two vertices are isolated and the third adds one unit edge. The first four vertices form a four-vertex path, of determinant 5. Appending vertices $4, 5, 6, 7$ applies the unit-leaf recurrence, giving $4, 3, 2, 1$ in turn. For F_4 the path edge products are $1, 2, 1$, so the recurrence gives $2, 3, 2, 1$. For G_2 the product 3 gives $4 - 3 = 1$.

For the exceptional matrices set $H = \mathrm{diag}(d)M$, with d as in (3). The following data give

$$H = L\,\mathrm{diag}(q)L^T, \qquad \mathbf{x}^T H \mathbf{x} = \sum_i q(i)\big((L^T\mathbf{x})(i)\big)^2. \tag{19}$$

In every case L is lower triangular with diagonal 1. All its entries strictly below the diagonal are zero except those specified here:

type	$(q(0), \ldots, q(\ell-1))$
E_8	$(2, 2, 3/2, 5/6, 4/5, 3/4, 2/3, 1/2)$
F_4	$(2, 3/2, 4/3, 1)$
G_2	$(2, 3/2)$.

For E_8 the nonzero lower entries are

$$L(2,0) = -\tfrac{1}{2}, \quad L(3,1) = -\tfrac{1}{2}, \quad L(3,2) = -\tfrac{2}{3}, \quad L(4,3) = -\tfrac{6}{5},$$
$$L(5,4) = -\tfrac{5}{4}, \quad L(6,5) = -\tfrac{4}{3}, \quad L(7,6) = -\tfrac{3}{2}.$$

The leading six or seven rows and columns of L, with the corresponding prefix of q, give the certificates for E_6 and E_7. For F_4 they are $L(1,0) = -1/2$, $L(2,1) = -4/3$, $L(3,2) = -3/2$. For G_2 the only lower entry is $L(1,0) = -3/2$. Multiplication verifies (19) entry by entry: each entry is $\sum_k L(i,k)q(k)L(j,k)$, with the rational data above. Since every $q(i) > 0$ and a triangular matrix with diagonal 1 is invertible, the displayed sum is positive for every nonzero $\mathbf{x}$. These are finite rational certificates, which we also check in `checks/affine.py`. For arbitrary classical rank we retain the positivity proof through Theorem 5.6; we do not use Sylvester's criterion as a Mathlib premise.

The base vectors and all their coefficients are the data in (5), the classical expansion formulas, (7)–(8), (10)–(11), and (13)–(14). For clarity, their squared simple lengths and Gram

determinants are

type	$(\langle\alpha_i,\alpha_i\rangle)_i$	$\det(\langle\alpha_i,\alpha_j\rangle)_{i,j}$
A_ℓ	$(2,\dots,2)$	$\ell+1$
B_ℓ	$(2,\dots,2,1)$	1
C_ℓ	$(2,\dots,2,4)$	4
D_ℓ	$(2,\dots,2)$	4
E_6, E_7, E_8	$(2,\dots,2)$	$3, 2, 1$ respectively
F_4	$(2,2,1,1)$	$1/4$
G_2	$(6,2)$	3.

Indeed the simple Gram matrix is $M\,\mathrm{diag}(\langle\alpha_i,\alpha_i\rangle/2)$, so its determinant follows from (17). This also confirms index by index that the short simple root in B is $\ell-1$, the short simple roots in C are $0,\dots,\ell-2$, in F_4 they are 2 and 3, and in G_2 the short simple root is 1.

A.3. **Root counts and line frames.** In Table 2, "short" and "long" are used only in the two-length cases. For a one-length system we put all roots in the single-length column. The factor t is N/n in (15).

Type	Single length	Short	Long	Lines N	t
A_ℓ	$\ell(\ell+1)$			$\ell(\ell+1)/2$	$(\ell+1)/2$
B_ℓ		2ℓ	$2\ell(\ell-1)$	ℓ^2	ℓ
C_ℓ		$2\ell(\ell-1)$	2ℓ	ℓ^2	ℓ
D_ℓ	$2\ell(\ell-1)$			$\ell(\ell-1)$	$\ell-1$
E_6	72			36	6
E_7	126			63	9
E_8	240			120	15
F_4		24	24	24	6
G_2		6	6	6	3

TABLE 2. Root and line counts in the ranges of Theorem B.

Here are coordinate frame calculations for each system in Remark 3.11. They provide checks separate from the reflection-invariance proof. For each pair $i<j$ the two normalized diagonal lines satisfy

$$\frac{(\mathbf{u}_i+\mathbf{u}_j)(\mathbf{u}_i+\mathbf{u}_j)^T}{2}+\frac{(\mathbf{u}_i-\mathbf{u}_j)(\mathbf{u}_i-\mathbf{u}_j)^T}{2}=\mathbf{u}_i\mathbf{u}_i^T+\mathbf{u}_j\mathbf{u}_j^T.$$

Summing over pairs gives $(n-1)I$ for the D_n lines. Adding the n axes gives nI for the B_n, C_n lines. For F_4 the 12 long lines contribute $3I_4$, the four axes contribute I_4, and the eight half-coordinate lines represented by

$$\tfrac{1}{2}(\mathbf{u}_1+\epsilon(2)\mathbf{u}_2+\epsilon(3)\mathbf{u}_3+\epsilon(4)\mathbf{u}_4),\qquad(\epsilon(2),\epsilon(3),\epsilon(4))\in\{1,-1\}^3,$$

contribute $2I_4$. Each diagonal entry in the latter sum is $8/4=2$; in an off-diagonal entry a sign that occurs to the first power can be flipped, pairing each summand with its negative. The total is $6I_4$.

For G_2 use the three short representatives $(\mathbf{u}_i-\mathbf{u}_j)/\sqrt{2}$ for $i<j$ and the three long representatives $(2\mathbf{u}_i-\mathbf{u}_j-\mathbf{u}_k)/\sqrt{6}$. Let $\phi=I_3-\mathbf{j}\mathbf{j}^T/3$, the orthogonal projection to the zero-sum plane. Expanding the short sum gives diagonal entries 1 and off-diagonal entries $-1/2$, namely $3\phi/2$. The ith long vector is $\sqrt{3/2}\,\phi\mathbf{u}_i$, so their sum of outer products is $(3/2)\phi\phi^T=(3/2)\phi$. The full sum is 3ϕ, or $3I$ on the plane.

For A_n, expansion of $\frac{1}{2}\sum_{i<j}(\mathbf{u}_i-\mathbf{u}_j)(\mathbf{u}_i-\mathbf{u}_j)^T$ gives $((n+1)I-\mathbf{j}\mathbf{j}^T)/2$, which acts as $(n+1)I/2$ on its zero-sum space. For the E systems, the root lists, their verified indecomposability and

Lemma 10.3 give exactly the factors $6, 9, 15$ on their spans. An exact coordinate alternative is the finite identity

$$\frac{1}{2}\sum_{\alpha\in\Phi}\frac{\alpha\alpha^T}{\langle\alpha,\alpha\rangle} = \frac{N}{n}\,A(A^TA)^{-1}A^T, \tag{20}$$

where A has the simple roots as columns. Here $A(A^TA)^{-1}A^T$ is the orthogonal projection onto the span: it fixes each column of A and annihilates their common orthogonal complement. Formula (20) involves only rational entries in every one of our models, including $\mathrm{E}_6, \mathrm{E}_7$ in their eight-coordinate ambient spaces. It is checked by the scripts.

A.4. **Checks and their scope.** The scripts need Python 3 with NumPy, SymPy and NetworkX. They can be run from this document's directory, as `python checks/models.py` and so on, for the following scripts:

`checks/models.py`
`checks/affine.py`
`checks/enumerate_cartan.py`
`checks/symbolic.py`
`checks/mathlib_literal.py`
`checks/trees_to_ten.py`
`checks/unit_trees.py`
`checks/matrix_equations.py`
`checks/alternative_reading.py`.

Each script has assertions and prints its results. They have the following separate roles.

(1) `models.py` uses rational coordinates to check every ordered pair of roots for crystallographic integrality and reflection closure, sign pairs and lines for reducedness and counts, all base coefficients for integrality and a common sign, the exact Cartan matrices and base Gram determinants, every pairwise squared cosine, and (20). The checked classical ranks are $1 \le \ell \le 8$ for A, $2 \le \ell \le 8$ for B, C, and $4 \le \ell \le 8$ for D. All three E systems and $\mathrm{F}_4, \mathrm{G}_2$ are checked completely. It also checks equality of the B, C line sets and the displayed B_2 similarity. The exceptional root counts are $72, 126, 240, 48, 12$, and the exact frame factors are $6, 9, 15, 6, 3$, as in Table 2.

(2) `affine.py` checks $B\mathbf{x} = 2\mathbf{x}$ and a one-dimensional kernel for all exceptional obstructions and the families through parameter 16. It checks the arm recurrence symbolically, all the leading-minor formulas through rank 16 in the classical cases, every exceptional leading minor, the symmetrizers, and all the small-rank permutations in Remark 6.1. It also checks the general leaf expansion at every leaf in those matrices and verifies all five rational certificates in (19) by exact matrix multiplication. The integer recurrences above prove the unlimited-rank statements.

(3) `enumerate_cartan.py` is an independent exhaustive check of all connected weighted supports on at most seven vertices. NetworkX's graph atlas supplies all 996 nonempty connected simple supports in this range. Exact rational elimination tests $2I - A$ for each unit support; 983 are not positive definite, and entrywise monotonicity excludes all their weighted refinements. All 13 remaining supports are trees, a fact the script checks rather than assumes. It tries all 3,145 assignments of squared edge weights in $\{1, 2, 3\}$ on these supports. For an edge directed from a parent to a child, take entries $-p, -1$ and propagate $d(\text{child}) = p\,d(\text{parent})$. Exact rational elimination of DM decides positivity, independently of any graph-type test. There are 26 accepted weighted assignments. Their 39 possible orientations, counting labellings that can be isomorphic, each match exactly one canonical matrix type in our restricted list, by directed graph isomorphism preserving the

entries. All types of rank at most 7 appear. Numerical largest eigenvalues from NumPy agree with every exact decision; numerical tolerances are not used for acceptance. This enumeration covers weighted graphs up to isomorphism, with possible repeated representatives, because every support is represented and every assignment on a retained support is tried. It does not prove the classification in unbounded rank.

(4) `symbolic.py` checks the coefficient inverses in (8) and (11) as symbolic identities, the three-line example, scalar identities for the determinant recurrences, and the pairwise and G_2 frame formulas. Its E_8 check is the arithmetic $2(120/8)-2=28$. The matrix determinant checks are in `affine.py`, and the matrix-equation checks are in `matrix_equations.py`.

(5) `mathlib_literal.py` compares the seven defining bodies with `Mathlib/LinearAlgebra/Matrix/Cartan.lean` at the pinned commit, ignoring whitespace only. It then compares an independent literal transcription with the shared `models.cartan` function through rank 16, including ranks 0 and 1, and checks the formulas of Section 6, Remark 6.1, and `CartanMatrix.E_det` through rank 14. This checks our Python transcription against the source; it does not execute Lean.

(6) `trees_to_ten.py` extends the weighted enumeration to all trees on 8, 9 and 10 vertices, respectively 23, 47 and 106 supports. It excludes 20, 45 and 104 at unit weights by exact leading determinants. On the surviving supports it tests all 6,561, 13,122 and 39,366 assignments, respectively, and compares every orientation of each accepted assignment with the literal Mathlib matrices. There are 5, 4 and 4 accepted assignments, and 7, 6 and 6 orientations. This includes E_8 and the tree obstructions on 8 and 9 vertices. We exclude non-tree supports in these ranks by the proved cycle obstruction and monotonicity, rather than enumerate them.

(7) `unit_trees.py` checks every tree through 14 vertices. The test for largest eigenvalue less than 2 is exact; the equality-to-2 comparison with the Smith list uses numerical tolerance 10^{-9}. For every tree satisfying the recognition hypotheses it also constructs and checks the path bijections, all possible prescribed endpoints, and the labelled arms. It checks the resulting D, E reindexings of Section 6 exactly.

(8) `matrix_equations.py` multiplies $(C+2I)(C-\theta I)$ directly in exact arithmetic, using reproducible choices of signs. It checks B_n, C_n for $2 \le n \le 5$, F_4, G_2, and the equations of Theorem 3.7 for A_n $(1 \le n \le 5)$, D_n $(4 \le n \le 6)$, and all three E systems. For E_8 it verifies the constant 28 and the nonzero residual $20G$ for the printed constant 18.

(9) `alternative_reading.py` compares unit representatives with actual unequal-length roots, checking the raw-root Gram factors $2n-1, 2n+2, 9, 12$ for B_n, C_n, F_4, G_2 at its specified ranks. Only the unit normalization yields the constants printed in Remark 3.11.

All counts and exact identities above passed. We use these Python computations as checks, not as mathematical premises or Lean proofs.

References

[CGSS] P. J. Cameron, J. M. Goethals, J. J. Seidel and E. E. Shult, *Line graphs, root systems, and elliptic geometry*, J. Algebra **43** (1976), 305–327.

[BCN] A. E. Brouwer, A. M. Cohen and A. Neumaier, *Distance-Regular Graphs*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3), vol. 18, Springer, 1989, Sections 3.1, 3.2 and 3.10.

[Mathlib] The mathlib community, *Mathlib*, commit `082e2d37` (4 September 2026; full identifier in the introduction), the version pinned in `lake-manifest.json`. The exact files, declarations and hypotheses used here are listed in Section 12.